FACULTY OF ENGINEERING • KHON KAEN UNIVERSITY

# Fundamental Electrical Engineering (FunE)

## Final Exam Master Flashcards & Solved Problems Booklet (72 Cards)

**COURSE & INSTRUCTOR**
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**COVERAGE**
Chapters 6, 7, 8, 9, 10 & Semiconductors

**CONTENT INCLUSIONS**
72 Cards • 22 Circuit Diagrams • Step-by-Step Solutions

PART 1

## Chapter 6: Three-Phase Circuits

Balanced & Unbalanced Y-Y, Y-Δ, Complex Power, Neutral Shift, 2-Wattmeter Method (10 Flashcards)

**CH06-01** Chapter 6 Level 1 (Core Concept) 📌 Lecture Slides & Formula Sheet

### Star (Y) vs Delta (Δ) Voltage & Current Relations

**QUESTION / EXAM PROBLEM:**

In a balanced 3-phase positive-sequence ($abc$) system, state the exact relationships between line quantities ($V_L, I_L$) and phase quantities ($V_p, I_p$) for both Y-connection and $\Delta$-connection.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Y: $V_L = \sqrt{3}V_p \angle +30^\circ, I_L = I_p$ Δ: $V_L = V_p, I_L = \sqrt{3}I_p \angle -30^\circ$

Step-by-Step Solution:

1. Star (Y) Connection: - Line voltage leads phase voltage by $30^\circ$: $V_{ab} = \sqrt{3}V_{an} \angle +30^\circ$ - Line current equals phase current: $I_L = I_p$ 2. Delta (Δ) Connection: - Line voltage equals phase voltage: $V_L = V_p$ - Line current lags phase current by $30^\circ$: $I_a = \sqrt{3}I_{ab} \angle -30^\circ$

Pitfall / Pro Tip:

In negative sequence (acb), line voltage leads by -30° (or lags by 30°). Remember: in Y, currents are identical; in Δ, voltages are identical.

**CH06-02** Chapter 6 Level 1 (Core Concept) 📌 HW Ch.6 Problem 1 & 5

### Delta to Wye (Δ ↔ Y) Impedance Conversion

**QUESTION / EXAM PROBLEM:**

If a balanced $3\phi$ load is connected in $\Delta$ with impedance $Z_\Delta = 60 + j30\ \Omega$, what is the equivalent impedance per phase when converted to a Wye (Y) connection ($Z_Y$)?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$Z_Y = \frac{Z_\Delta}{3}$

Step-by-Step Solution:

$$Z_Y = \frac{Z_\Delta}{3} = \frac{60 + j30}{3} = 20 + j10\ \Omega$$

Conversely, if converting from Wye to Delta: $Z_\Delta = 3Z_Y$.

Pitfall / Pro Tip:

Multiplying by 3 instead of dividing is a classic mistake! A Delta load draws 3 times more current than a Wye load of the same impedance value, so equivalent Wye impedance must be 3 times smaller.

**CH06-03** Chapter 6 Level 1 (Core Concept) Formula Sheet & Lecture Slides

## Three-Phase Power Formulas & Complex Power

**QUESTION / EXAM PROBLEM:**

State the formulas for balanced $3\phi$ Real Power ($P_{3\phi}$), Reactive Power ($Q_{3\phi}$), Apparent Power ($|S_{3\phi}|$), and Complex Power ($S_{3\phi}$) in terms of line values ($V_L, I_L$).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$P_{3\phi} = \sqrt{3}V_L I_L \cos\theta \quad [\text{W}]\ Q_{3\phi} = \sqrt{3}V_L I_L \sin\theta \quad [\text{VAR}]\ |S_{3\phi}| = \sqrt{3}V_L I_L \quad [\text{VA}]\ S_{3\phi} = \sqrt{3}V_L I_L^* = P + jQ$

Step-by-Step Solution:

When using Line values ($V_L, I_L$), always include the factor $\sqrt{3}$. When using Phase values ($V_p, I_p$), use factor 3: - $P = 3V_p I_p \cos\theta$ - $Q = 3V_p I_p \sin\theta$ - $S = 3V_p I_p^*$ where $\theta = \angle Z_p$ is the load phase impedance angle, and $PF = \cos\theta$.

Pitfall / Pro Tip:

Never mix $\sqrt{3}$ with phase quantities! It is either $\sqrt{3}V_L I_L$ or $3V_p I_p$.

**CH06-04** Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 1 (Page 1)

## Balanced 4-Wire Y-Y System Full Calculation

**QUESTION / EXAM PROBLEM:**

In the balanced Y-Y circuit shown below, the source phase voltages are $V_{an} = 220\angle 0^\circ$ V, $V_{bn} = 220\angle -120^\circ$ V, $V_{cn} = 220\angle -240^\circ$ V. Line impedance is $Z_{\text{line}} = 10 + j5\ \Omega$. Balanced load impedance is $Z_L = 20 + j50\ \Omega$. Find: (1) Line voltages $V_{ab}, V_{bc}, V_{ca}$, (2) Line currents $I_{aA}, I_{bB}, I_{cC}$, (3) Neutral current $I_{Nn}$, (4) Total $3\phi$ complex power $S$.

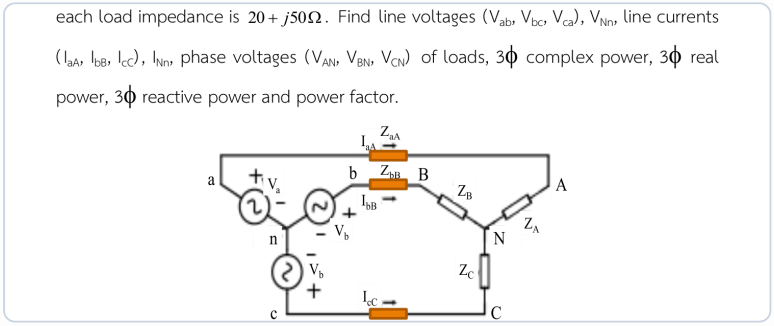



Fig. 1: Balanced Y-Y System with Line Impedance

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_{ab} = \sqrt{3}V_{an}\angle +30^\circ\ I_{aA} = \frac{V_{an}}{Z_{\text{line}}+Z_L}\ I_{Nn} = 0\ S_{3\phi} = 3V_{an}I_{aA}^*$

Step-by-Step Solution:

1. Line Voltages: - $V_{ab} = \sqrt{3}(220)\angle(0^\circ + 30^\circ) = 381.05\angle 30^\circ$ V - $V_{bc} = 381.05\angle -90^\circ$ V - $V_{ca} = 381.05\angle -210^\circ$ V 2. Per-phase total impedance: $Z_{\text{total}} = (10 + j5) + (20 + j50) = 30 + j55\ \Omega = 62.65\angle 61.39^\circ\ \Omega$ 3. Line Currents: - $I_{aA} = \frac{220\angle 0^\circ}{62.65\angle 61.39^\circ} = 3.51\angle -61.39^\circ$ A - $I_{bB} = 3.51\angle -181.39^\circ$ A - $I_{cC} = 3.51\angle 58.61^\circ$ A 4. Neutral Current: In a balanced system, $I_{Nn} = -(I_{aA} + I_{bB} + I_{cC}) = 0$ A. $V_{Nn} = 0$ V. 5. Total $3\phi$ Power: $S = 3 \times 220 \times 3.51\angle +61.39^\circ = 2,316.6\angle 61.39^\circ$ VA $= 1,109$ W $+ j2,034$ VAR.

Pitfall / Pro Tip:

In a balanced system, neutral voltage shift $V_{Nn}$ is strictly ZERO, and neutral current $I_{Nn} = 0$.

CH06-05 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 2 (Page 2)

## Unbalanced 3-Wire Y-Y System (Neutral Shift)

**QUESTION / EXAM PROBLEM:**

In the 3-wire Y-Y system below (NO neutral wire), source is 220 V balanced, $Z_{\text{line}} = 10 + j5\ \Omega$, but load is UNBALANCED: $Z_A = 20 + j50\ \Omega$, $Z_B = 10 + j50\ \Omega$, $Z_C = 20 + j20\ \Omega$. How is the neutral voltage shift $V_{Nn}$ calculated, and why do load phase voltages become unbalanced?

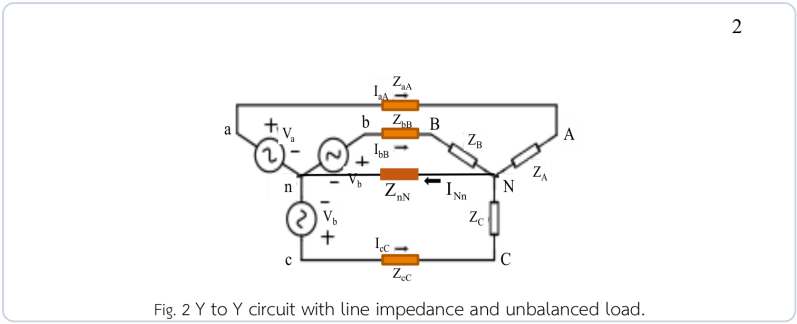


Fig. 2 Y to Y circuit with line impedance and unbalanced load.

Fig. 2: Unbalanced 3-Wire Y-Y System without Neutral

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C}$ (Millman's Theorem)

Step-by-Step Solution:

1. Millman's Theorem for Neutral Shift: Define total branch admittance $Y_A = 1/(Z_{\text{line}} + Z_A)$, $Y_B = 1/(Z_{\text{line}} + Z_B)$, $Y_C = 1/(Z_{\text{line}} + Z_C)$.

$$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C} \neq 0$$

2. Line Currents with Neutral Shift: - $I_{aA} = (V_{an} - V_{Nn})Y_A$ - $I_{bB} = (V_{bn} - V_{Nn})Y_B$ - $I_{cC} = (V_{cn} - V_{Nn})Y_C$ Since no neutral wire exists, $I_{aA} + I_{bB} + I_{cC} = 0$. 3. Impact on Load Phase Voltages: Load phase voltage $V_{AN} = V_{an} - I_{aA}Z_{\text{line}} - V_{Nn}$. Because $V_{Nn} \neq 0$, phase voltages across the load become severely distorted and unbalanced.

Pitfall / Pro Tip:

Without a neutral wire, the neutral point floats ($V_{Nn} \neq 0$), causing dangerous overvoltage on lighter-loaded phases!

CH06-06 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 3 (Page 2)

## Unbalanced Y-Y System with Neutral Line Impedance

**QUESTION / EXAM PROBLEM:**

In the 4-wire Y-Y system below, an unbalanced load ($Z_A \neq Z_B \neq Z_C$) is connected with a neutral return wire having finite impedance $Z_{nN}$. Write the exact node-voltage equation to solve for $V_{Nn}$ and neutral current $I_{Nn}$.

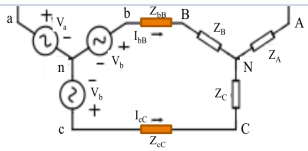



Fig. 3 Y to Y circuit with line impedance and unbalanced load.

Adapted from: https://en.wikipedia.org/wiki/Three-phase_electric_power

Fig. 3: Unbalanced 4-Wire Y-Y System with Neutral Impedance

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_{Nn} = \frac{V_{an}Y_A + V_{bn}Y_B + V_{cn}Y_C}{Y_A + Y_B + Y_C + Y_n}$ $I_{Nn} = V_{Nn}Y_n = \frac{V_{Nn}}{Z_{nN}}$

Step-by-Step Solution:

1. Node-Voltage Equation at node N: By KCL at load neutral node N:

$$\frac{V_{Nn} - V_{an}}{Z_A + Z_{\text{line}}} + \frac{V_{Nn} - V_{bn}}{Z_B + Z_{\text{line}}} + \frac{V_{Nn} - V_{cn}}{Z_C + Z_{\text{line}}} + \frac{V_{Nn}}{Z_{nN}} = 0$$

2. Solving for $V_{Nn}$:

$$V_{Nn} = \frac{V_{an}Y'_A + V_{bn}Y'_B + V_{cn}Y'_C}{Y'_A + Y'_B + Y'_C + Y_n}$$

where $Y'_k = 1/(Z_k + Z_{\text{line}})$ and $Y_n = 1/Z_{nN}$. 3. Neutral Return Current:

$$I_{Nn} = \frac{V_{Nn}}{Z_{nN}} = -(I_{aA} + I_{bB} + I_{cC})$$

.

Pitfall / Pro Tip:

Do not omit $Y_n = 1/Z_{nN}$ in the denominator of Millman's formula! As $Z_{nN} \to 0$, $V_{Nn} \to 0$ (solid ground).

CH06-07 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 4 (Page 3)

## Y-Connected Induction Motor Load Calculation

**QUESTION / EXAM PROBLEM:**

A $3\phi$, 12-hp induction motor is Y-connected to a $3\phi$ supply with line-to-line voltage $V_L = 380\ \text{V}, 50\ \text{Hz}$. The motor has $\eta = 90\%$ efficiency and a lagging power factor of 0.85. Calculate: (1) Input real power ($P_{\text{in}}$), (2) Line current ($I_L$), (3) Per-phase impedance magnitude ($|Z|$), resistance ($R$), and inductance ($L$). ($1\ \text{hp} = 746\ \text{W}$)

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$P_{\text{out}} = 12 \times 746\ \text{W}\ P_{\text{in}} = P_{\text{out}}/\eta\ I_L = \frac{P_{\text{in}}}{\sqrt{3}V_L \cos\theta}\ V_p = V_L/\sqrt{3}\ Z = V_p/I_p$

Step-by-Step Solution:

1. Input Power: $P_{\text{out}} = 12 \times 746 = 8,952\ \text{W}\ P_{\text{in}} = 8,952/0.90 = 9,946.67\ \text{W}\ (9.95\ \text{kW})$ 2. Line Current:

$$I_L = \frac{9,946.67}{\sqrt{3} \times 380 \times 0.85} = 17.78\ \text{A}$$

3. Phase Impedance (Y-connection: $V_p = 380/\sqrt{3} = 219.39\ \text{V}, I_p = I_L = 17.78\ \text{A}$): - $|Z| = 219.39/17.78 = 12.34\ \Omega$ - $\theta = \arccos(0.85) = 31.79°$ - $Z = 12.34\angle 31.79° = 10.49 + j6.50\ \Omega$ - Resistance: $R = 10.49\ \Omega$ - Inductance: $X_L = 2\pi f L = 6.50 \implies L = 6.50/(2\pi \times 50) = 20.7\ \text{mH}$.

Pitfall / Pro Tip:

Always convert output mechanical horsepower to electrical input power using efficiency ($P_{\text{in}} = P_{\text{out}}/\eta$) BEFORE calculating current!

CH06-08 Chapter 6 Level 2 (Past Exam / Calc) HW Ch.6 Problem 5 & Lecture Slides

## Delta-Connected Motor Load Calculation

**QUESTION / EXAM PROBLEM:**

A $3\phi$, 10-hp motor is $\Delta$-connected to a $3\phi$ supply with $V_L = 380\ \text{V}, 50\ \text{Hz}$, $\eta = 85\%$, and $PF = 0.8$ lagging. Find: (1) Line current ($I_L$), (2) Phase current ($I_p$), (3) Per-phase $\Delta$ impedance ($Z_\Delta$).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$P_{\text{in}} = (10 \times 746)/0.85 = 8,776.47\ \text{W}\ I_L = \frac{P_{\text{in}}}{\sqrt{3}V_L \cos\theta}\ I_p = I_L/\sqrt{3}, V_p = V_L = 380\ \text{V}\ Z_\Delta = V_p/I_p$

Step-by-Step Solution:

1. Input Power: $P_{\text{in}} = 7,460/0.85 = 8,776.47\ \text{W}$ 2. Line Current:

$$I_L = \frac{8,776.47}{\sqrt{3} \times 380 \times 0.8} = 16.67\ \text{A}$$

3. Phase Quantities & Impedance: - Phase current: $I_p = I_L/\sqrt{3} = 16.67/\sqrt{3} = 9.62\ \text{A}$ - Phase voltage: $V_p = V_L = 380\ \text{V}$ - Impedance magnitude: $|Z_\Delta| = 380/9.62 = 39.5\ \Omega$ - Angle: $\theta = \arccos(0.8) = 36.87°$ - $Z_\Delta = 39.5\angle 36.87° = 31.6 + j23.7\ \Omega$ ($R_\Delta = 31.6\ \Omega, X_{L\Delta} = 23.7\ \Omega$).

Pitfall / Pro Tip:

In Delta, $V_p = V_L$, but $I_p = I_L/\sqrt{3}$. Do not divide voltage by $\sqrt{3}$!

**CH06-09** Chapter 6 Level 2 (Past Exam / Calc) 📌 Lecture Slides & Jele Formula Sheet

## Two-Wattmeter Method for 3-Phase Power Measurement

**QUESTION / EXAM PROBLEM:**

Explain the Two-Wattmeter Method for measuring balanced 3-phase power. State the formulas for total active power ($P$), reactive power ($Q$), and power factor angle ($\theta$) from meter readings $W_1$ and $W_2$.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$P_{\text{total}} = W_1 + W_2$ $Q_{\text{total}} = \sqrt{3}(W_1 - W_2)$ $\tan\theta = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$

Step-by-Step Solution:

1. Total Real Power: $P = W_1 + W_2$ 2. Total Reactive Power: $Q = \sqrt{3}(W_1 - W_2)$ 3. Power Factor Angle: $\theta = \arctan\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)$ 4. Power Factor: $PF = \cos\theta$ Special Cases: - If $W_1 = W_2$: $\theta = 0^\circ \implies PF = 1.0$ (purely resistive). - If $W_2 = 0$: $PF = 0.5$ ( $theta = 60^\circ$). - If $W_2 < 0$: $PF < 0.5$ (one wattmeter reading turns negative).

Pitfall / Pro Tip:

If $PF < 0.5$, one wattmeter deflects backward. In calculations, you MUST preserve the negative sign (e.g. $W_1 + (-W_2)$).

**CH06-10** Chapter 6 Level 1 (Core Concept) 📌 HW Ch.6 Problem 1, 2, 3

## Neutral Wire Current in Balanced vs Unbalanced Systems

**QUESTION / EXAM PROBLEM:**

What is the neutral current $I_N$ in a balanced 4-wire Y-Y system? What causes current to flow in the neutral wire in an unbalanced system?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_N = -(I_a + I_b + I_c)$

Step-by-Step Solution:

1. Balanced System: $I_a + I_b + I_c = 0$, therefore neutral current is strictly ZERO ($I_N = 0$). No current returns through the neutral line. 2. Unbalanced System: The phase currents do not sum to zero due to unequal load impedances ($Z_A \neq Z_B \neq Z_C$), so neutral current $I_N = -(I_a + I_b + I_c) \neq 0$ carries the return unbalance current.

Pitfall / Pro Tip:

If the neutral wire is disconnected in an unbalanced load, the neutral point shifts ($V_{Nn} \neq 0$), causing severe overvoltage across the lighter load phase!

PART 2

# Chapter 7: Electrical Machines (DC & AC)

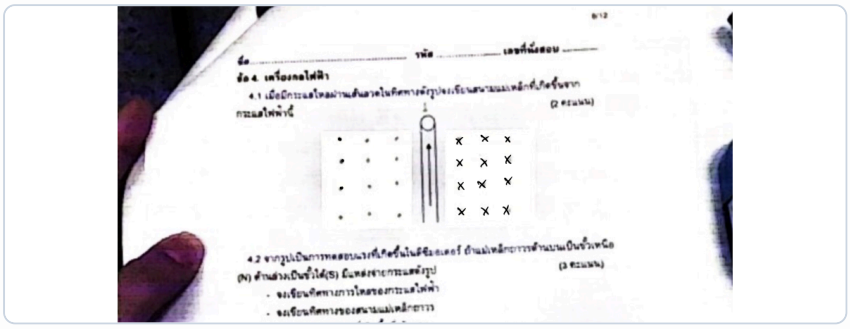
Right-Hand Rule, Lorentz Force, DC Shunt/Series Motors & Generators, Induction & Synchronous (15 Flashcards)

**CH07-01** Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.1 (Page 28 - 2 Points)

## Right-Hand Grip Rule for Magnetic Field around a Conductor

**QUESTION / EXAM PROBLEM:**

Current flows upward along a straight vertical wire. Using the right-hand grip rule, sketch/state the direction of the magnetic field on the left and right sides of the wire.



Exam Problem 4.1: Magnetic Field around Current Conductor

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Right-Hand Grip Rule: Thumb = Current ($I$), Fingers = Field ($B$)

Step-by-Step Solution:

Point right thumb upward along the current flow: - Left side of the wire: Magnetic field lines point OUT OF THE PAGE (represented by dots •). - Right side of the wire: Magnetic field lines point INTO THE PAGE (represented by crosses ×).

Pitfall / Pro Tip:

Direct 2-point exam question! Remember: Dot (•) = coming towards you / out of paper; Cross (×) = going away from you / into paper.

**CH07-02** Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.2 (Page 28 - 4 Points)

## Lorentz Force on Current-Carrying Conductor in DC Motor

**QUESTION / EXAM PROBLEM:**

State the formula and rule used to determine the electromagnetic force developed on a conductor in a DC motor armature.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$\vec{F} = I(\vec{L} \times \vec{B}) \implies F = BIL\sin\alpha$

Step-by-Step Solution:

1. Formula: $F = BIL\sin\alpha$, where $B$ is flux density (Tesla), $I$ is conductor current (A), $L$ is active conductor length (m), and $\alpha$ is the angle between wire and magnetic field (usually $90°$, so $F = BIL$). 2. Direction: Determined by Fleming's Left-Hand Rule for Motors: - Thumb = Motion / Force ($F$) - First finger = Magnetic Field ($B$, North to South) - Second finger = Current ($I$, positive to negative).

Pitfall / Pro Tip:

Use LEFT hand for motors (converting electrical to mechanical motion); use RIGHT hand for generators (inducing EMF).

CH07-03 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.3 (Page 28 - 4 Points)

## DC Machine Stator and Rotor Structural Components

**QUESTION / EXAM PROBLEM:**

What are the two main structural parts of a DC machine? Name at least 2 components found in each part.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Machine = Stator (Stationary) + Rotor (Rotating)

Step-by-Step Solution:

1. Stator (Stationary Part): - Frame / Yoke (cast iron outer housing and magnetic return path) - Main Field Poles & Pole Shoes (laminated sheet steel) - Field Windings (copper coils producing main flux $\Phi$) - End covers and brush holders. 2. Rotor / Armature (Rotating Part): - Armature Core (laminated silicon steel with slots) - Armature Windings (copper coils carrying load current) - Commutator (segmented copper bars with mica insulation) - Shaft, bearings, and cooling fan.

Pitfall / Pro Tip:

Armature core is made of laminated silicon steel sheets specifically to minimize Eddy current losses and Hysteresis losses.

CH07-04 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

## DC Machine Back EMF Formula & Physical Meaning

**QUESTION / EXAM PROBLEM:**

State the equation for induced armature back-EMF ($E_a$) in a DC machine and define all variables. How does it relate to motor speed?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$E_a = K_a \Phi \omega_m = \left(\frac{pZ}{2\pi a}\right) \Phi \omega_m$

Step-by-Step Solution:

- $E_a$ = Induced armature back EMF (V) - $K_a = \frac{pZ}{2\pi a}$ = Machine design constant ($p$ = poles, $Z$ = total conductors, $a$ = parallel paths) - $\Phi$ = Magnetic flux per pole (Wb) - $\omega_m$ = Mechanical angular velocity (rad/s) = $\frac{2\pi n}{60}$ ($n$ in rpm). Physical Meaning: In a motor, $E_a$ opposes applied terminal voltage ($V_t$). As speed increases, $E_a$ increases, limiting armature current $I_a = (V_t - E_a)/R_a$.

Pitfall / Pro Tip:

At starting ($n = 0$), $E_a = 0$ V. Without starting resistors, inrush current $I_{\text{start}} = V_t/R_a$ can reach 10-20 times rated current!

CH07-05 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Homework 7

## DC Machine Developed Torque & Converted Power

**QUESTION / EXAM PROBLEM:**

State the formula for developed electromagnetic torque ($T_d$) in a DC motor and the relationship between developed mechanical power and electrical power.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$T_d = K_a \Phi I_a$ $P_{\text{developed}} = E_a I_a = T_d \omega_m$

Step-by-Step Solution:

1. Developed Torque: $T_d = K_a \Phi I_a$ [N · m]. It is strictly proportional to field flux and armature current. 2. Converted Power: $P_d = E_a I_a = T_d \omega_m$ [W]. The internal electrical power converted equals mechanical power developed. 3. Shaft Output Power: $P_{\text{shaft}} = P_d - P_{\text{rotational losses}}$.

Pitfall / Pro Tip:

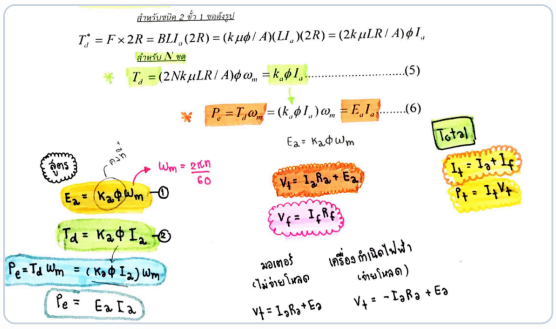
Do not confuse developed torque ($T_d$) with shaft torque ($T_{\text{shaft}}$). Rotational losses (friction + windage) reduce output torque.

**CH07-06** Chapter 7 Level 1 (Core Concept) FunE-final.pdf Page 3 & Lecture Slides

## DC Shunt Motor Equivalent Circuit & KVL Equations

**QUESTION / EXAM PROBLEM:**

Draw the equivalent circuit of a DC Shunt Motor and state the KVL voltage loop equation and KCL current node equation.




DC Shunt Motor Equivalent Circuit & KVL

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_t = E_a + I_a R_a \; I_t = I_a + I_f \; I_f = V_t / R_f$

Step-by-Step Solution:

1. Field Circuit (Shunt across terminals): $I_f = \frac{V_t}{R_f}$ 2. Armature Circuit (KVL):

$$V_t = E_a + I_a R_a \implies E_a = V_t - I_a R_a$$

3. Current Relation (KCL at input terminal):

$$I_t = I_a + I_f \implies I_a = I_t - I_f$$

4. Converted Mechanical Power:

$$P_{\text{conv}} = E_a I_a$$

.

Pitfall / Pro Tip:

In a MOTOR, $V_t > E_a$ because the source drives current into the machine ($V_t = E_a + I_a R_a$).

CH07-07 Chapter 7 Level 2 (Past Exam / Calc) FunE-final.pdf Page 3 (Example 1)

## DC Shunt Motor Solved Example (3 hp, 120 V)

**QUESTION / EXAM PROBLEM:**

A $3\ \text{hp}, 120\ \text{V}$ DC shunt motor has armature resistance $R_a = 0.2\ \Omega$, field resistance $R_f = 60\ \Omega$, and draws a full-load line current $I_t = 24.5\ \text{A}$. Calculate: (1) Field current $I_f$, (2) Armature current $I_a$, (3) Back EMF $E_a$, (4) Converted mechanical power $P_{\text{conv}}$, (5) Rotational loss if output is rated at $3\ \text{hp}$. ($1\ \text{hp} = 746\ \text{W}$)

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_f = V_t/R_f\ I_a = I_t - I_f\ E_a = V_t - I_aR_a\ P_{\text{conv}} = E_aI_a\ P_{\text{rot}} = P_{\text{conv}} - P_{\text{out}}$

Step-by-Step Solution:

1. Field Current:

$$I_f = \frac{120\ \text{V}}{60\ \Omega} = 2.0\ \text{A}$$

2. Armature Current:

$$I_a = I_t - I_f = 24.5 - 2.0 = 22.5\ \text{A}$$

3. Back EMF:

$$E_a = V_t - I_aR_a = 120 - (22.5 \times 0.2) = 120 - 4.5 = 115.5\ \text{V}$$

4. Converted Power:

$$P_{\text{conv}} = E_aI_a = 115.5\ \text{V} \times 22.5\ \text{A} = 2,598.75\ \text{W}$$

5. Rotational Loss: $P_{\text{out}} = 3 \times 746 = 2,238\ \text{W}$

$$P_{\text{rot}} = P_{\text{conv}} - P_{\text{out}} = 2,598.75 - 2,238 = 360.75\ \text{W}$$

.

Pitfall / Pro Tip:

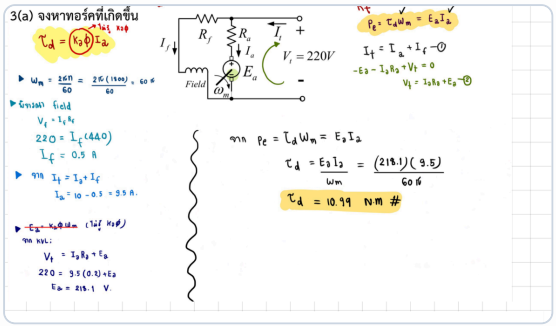
Always separate internal converted power ($P_{conv} = E_aI_a$) from output shaft power ($P_{out}$). The difference is rotational friction and windage loss.

**CH07-08** Chapter 7 🔴 Level 2 (Past Exam / Calc) 📌 FunE-final.pdf Page 5 (Example 3a & 3b)

## DC Shunt Motor Speed Control & Series Resistance

**QUESTION / EXAM PROBLEM:**

A 220 V DC shunt motor has $R_a = 0.5\ \Omega, R_f = 110\ \Omega$. At full load, $I_t = 42$ A and speed $n = 1,200$ rpm. (1) Find developed torque $T_d$ and constant $K_a\Phi$. (2) If series resistor $R_{\text{adj}} = 1.0\ \Omega$ is added in the armature branch and torque drops to $T_d' = 20\ \text{N}\cdot\text{m}$, calculate the new speed $n'$.




DC Shunt Motor Speed Control with Added Resistance

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_f = V_t/R_f, I_a = I_t - I_f\ E_a = V_t - I_aR_a\ \omega_m = \frac{2\pi n}{60}, K_a\Phi = E_a/\omega_m\ I_a' = T_d'/(K_a\Phi)\ E_a' = V_t - I_a'(R_a + R_{\text{adj}})\ n' = \frac{60E_a'}{2\pi(K_a\Phi)}$

Step-by-Step Solution:

1. Initial Condition: - $I_f = 220/110 = 2.0$ A - $I_a = 42 - 2 = 40.0$ A - $E_a = 220 - (40 \times 0.5) = 200.0$ V - $\omega_m = \frac{2\pi(1200)}{60} = 125.66$ rad/s - $K_a\Phi = \frac{E_a}{\omega_m} = \frac{200}{125.66} = 1.592\ \text{V}\cdot\text{s/rad}$ - $T_d = K_a\Phi I_a = 1.592 \times 40 = 63.66\ \text{N}\cdot\text{m}$ 2. With $R_{\text{adj}} = 1.0\ \Omega$ and $T_d' = 20\ \text{N}\cdot\text{m}$: - New armature current: $I_a' = \frac{20}{1.592} = 12.56$ A - Total armature resistance: $R_{\text{tot}} = 0.5 + 1.0 = 1.5\ \Omega$ - New Back EMF: $E_a' = 220 - (12.56 \times 1.5) = 220 - 18.84 = 201.16$ V - New Speed: $\omega_m' = \frac{201.16}{1.592} = 126.36$ rad/s - In RPM: $n' = \frac{60 \times 126.36}{2\pi} = 1,206.6$ rpm.

Pitfall / Pro Tip:

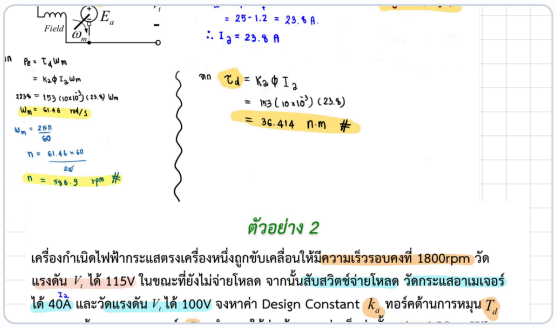
When $R_{adj}$ is inserted in the armature branch, total resistance becomes $R_a + R_{adj}$. Notice how reducing torque requirement allows speed to remain high even with added resistance.

**CH07-09** Chapter 7 Level 2 (Past Exam / Calc) FunE-final.pdf Page 4 (Example 2)

## DC Shunt Generator KVL, No-Load to Full-Load

**QUESTION / EXAM PROBLEM:**

A DC shunt generator driven at $1,800$ rpm produces terminal voltage $V_t = 115$ V at no load ($I_a = 0$). When loaded, it delivers $I_a = 40$ A at $V_t = 100$ V. Flux per pole is $\Phi = 150$ mWb. Find: (1) Machine constant $K_a$, (2) Armature resistance $R_a$, (3) Developed torque $T_d$.




DC Generator Circuit and KVL

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

No load: $E_a = V_t = 115$ V $\omega_m = \frac{2\pi n}{60} = 188.5$ rad/s $K_a = \frac{E_a}{\Phi \omega_m}$ Loaded: $E_a = V_t + I_a R_a \implies R_a = \frac{E_a - V_t}{I_a}$ $T_d = K_a \Phi I_a$

Step-by-Step Solution:

1. Angular speed: $\omega_m = \frac{2\pi \times 1800}{60} = 188.5$ rad/s 2. At no load ($I_a = 0$), $E_a = V_t = 115$ V:

$$K_a = \frac{115}{150 \times 10^{-3} \times 188.5} = \frac{115}{28.27} = 4.07$$

3. Loaded operation: Assuming speed and flux remain constant, $E_a = 115$ V.

$$R_a = \frac{E_a - V_t}{I_a} = \frac{115 - 100}{40} = \frac{15}{40} = 0.375\ \Omega$$

4. Developed Torque:

$$T_d = K_a \Phi I_a = 4.07 \times (150 \times 10^{-3}) \times 40 = 24.42\ \text{N} \cdot \text{m}$$

.

Pitfall / Pro Tip:

Crucial distinction: In a GENERATOR: $E_a = V_t + I_a R_a$ ($E_a > V_t$). In a MOTOR: $E_a = V_t - I_a R_a$ ($V_t > E_a$).

**CH07-10** Chapter 7 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.7 Problem 4 & Talui Homework Page 1

## DC Shunt Motor Equivalent Circuit & Speed/Torque Calculation

**QUESTION / EXAM PROBLEM:**

A 2-pole, 2.5 hp, 100 V DC shunt motor has $K_a = 153$, full-load total current $I_t = 25.0$ A, armature resistance $R_a = 0.75\ \Omega$, field flux $\Phi = 10.0$ mWb, and field resistance $R_f = 440\ \Omega$. Calculate: (1) Armature current ($I_a$), (2) Back EMF ($E_a$), (3) Motor speed ($n$ in rpm), (4) Developed torque ($T_d$).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_f = V_t/R_f\ I_a = I_t - I_f\ E_a = V_t - I_aR_a\ \omega_m = E_a/(K_a\Phi) \implies n = \frac{60\omega_m}{2\pi}\ T_d = K_a\Phi I_a$

Step-by-Step Solution:

1. Field current: $I_f = 100/440 = 0.23$ A 2. Armature current: $I_a = I_t - I_f = 25.0 - 0.23 = 24.77$ A 3. Back EMF: $E_a = V_t - I_aR_a = 100 - (24.77 \times 0.75) = 100 - 18.58 = 81.42$ V 4. Speed calculation:

$$E_a = K_a\Phi\omega_m = 153 \times (10 \times 10^{-3}) \times \left(\frac{2\pi n}{60}\right)$$

$$81.42 = 1.53 \times 0.10472 \times n = 0.16022 \times n \implies n = 508.17\ \text{rpm}$$

5. Developed Torque:

$$T_d = K_a\Phi I_a = 153 \times (10 \times 10^{-3}) \times 24.77 = 37.9\ \text{N} \cdot \text{m}$$

.

Pitfall / Pro Tip:

Always subtract field current $I_f$ from total supply current $I_t$ before computing $I_aR_a$ drop! Field current does not flow through armature.

**CH07-11** Chapter 7 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.7 Problem 5 & Talui Homework Page 2-3

## DC Series Motor Comparison & Runaway Hazard

**QUESTION / EXAM PROBLEM:**

A 2.5 hp series motor has $K_a = 153, R_a = 0.75\ \Omega, \Phi = 10.0$ mWb, $R_f = 440\ \Omega$. Find the required terminal voltages: (a) $V_{t1}$ to give the same speed (508.17 rpm) as Problem 4, (b) $V_{t2}$ to give the same torque (37.9 N · m) as Problem 4.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_t = I_a(R_a + R_f) + E_a$ (a) Same speed: $E_a = 81.42$ V, $I_a = P/E_a = 1865/81.42 = 22.91$ A (b) Same torque: $I_a = 24.77$ A, $E_a = P/I_a = 1865/24.77 = 75.3$ V

Step-by-Step Solution:

(a) For same speed ($n = 508.17$ rpm): $E_a = 81.42$ V $I_a = P_e/E_a = 1865/81.42 = 22.91$ A

$$V_{t1} = I_a(R_f + R_a) + E_a = 22.91(440 + 0.75) + 81.42 = 10,181\ \text{V} = 10.18\ \text{kV}$$

(b) For same torque ($T_d = 37.9$ N · m): $I_a = 24.77$ A $E_a = P_e/I_a = 1865/24.77 = 75.3$ V

$$V_{t2} = I_a(R_f + R_a) + E_a = 24.77(440.75) + 75.3 = 10,993\ \text{V} \approx 11.0\ \text{kV}$$

(c) Can same speed and same torque happen simultaneously? NO, because speed and torque uniquely dictate $E_a$ and $I_a$.

Pitfall / Pro Tip:

Why must a DC series motor never run at no load? Because $I_a \to 0 \implies \Phi \to 0 \implies \omega_m \approx V_t/(K_a\Phi) \to \infty$ (motor dangerously overspeeds and tears itself apart).

CH07-12 Chapter 7 Level 1 (Core Concept) 📌 Final Exam Problem 4.4 (Page 27 - 8 Points)

## Squirrel Cage Induction Motor & Reversing Rotation

**QUESTION / EXAM PROBLEM:**

What is a squirrel-cage rotor in a 3-phase induction motor? Explain its operating principle and describe how to reverse its direction of rotation.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Swap any 2 supply lines (Phase A and Phase B)

Step-by-Step Solution:

1. Structure: Rotor composed of thick copper or aluminum conductor bars embedded in a laminated steel core, permanently short-circuited at both ends by heavy conductive end-rings (resembling a squirrel exercise wheel). 2. Operating Principle: 3-phase AC in stator windings produces a Rotating Magnetic Field (RMF). The RMF sweeps across the rotor bars, inducing an EMF and heavy circulating currents. Interaction between rotor currents and the RMF produces driving torque following Lenz's Law. 3. Reversing Direction: Swap ANY TWO of the three incoming supply lines (e.g. swap A and B). This reverses the phase sequence from A-B-C to A-C-B, reversing the RMF rotation direction.

Pitfall / Pro Tip:

Direct 8-point exam question! Never say 'reverse DC polarity' because this is an AC motor without brushes or commutators.

CH07-13 Chapter 7 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

## Synchronous Speed, Rotor Speed, and Slip

**QUESTION / EXAM PROBLEM:**

Define Synchronous Speed ($N_s$) and Slip ($s$) in a 3-phase induction motor. Why can an induction motor never run at synchronous speed under normal motoring conditions?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$N_s = \frac{120f}{p} \quad [\text{rpm}]$ $s = \frac{N_s - N_r}{N_s} = \frac{\omega_s - \omega_r}{\omega_s}$

Step-by-Step Solution:

1. Synchronous Speed: $N_s = \frac{120f}{p}$ is the rotational speed of the stator magnetic field ($f$ = grid frequency in Hz, $p$ = total poles). 2. Slip: $s = \frac{N_s - N_r}{N_s}$, typically $2\% - 5\%$ at rated full load. 3. Why $N_r < N_s$: If the rotor ran at synchronous speed ($N_r = N_s$), the relative motion between rotor bars and magnetic field would become zero ($d\Phi/dt = 0$). No EMF would be induced, zero rotor current would flow, and developed torque would drop to zero. The rotor must slip behind $N_s$ to produce torque.

Pitfall / Pro Tip:

If $p$ is given as pole pairs $P$, the formula is $N_s = \frac{60f}{P}$. Always check whether the question specifies poles or pole pairs.

CH07-14 Chapter 7 Level 1 (Core Concept) 📌 AC Machines Slides P.12-14

## Types of Single-Phase Induction Motors

**QUESTION / EXAM PROBLEM:**

Single-phase induction motors are not self-starting. Name the 3 common types used to create starting torque and describe their operating features.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Shaded-Pole | Capacitor-Start | Universal Motor

Step-by-Step Solution:

1. Shaded-Pole Motor: - Uses copper shading rings on pole faces to create a delayed phase shift in magnetic flux. - Low starting torque, simple, low cost; used in small desk fans and blowers. 2. Capacitor-Start Motor: - Uses an auxiliary starting winding in series with a capacitor and centrifugal switch. - High starting torque ($200-350\%$ of rated); switch disconnects capacitor at $\approx 75\%$ speed; used in air compressors, water pumps, refrigerators. 3. Universal Motor: - Series-wound motor that operates on both AC and DC. - Very high speed (up to 20,000 rpm) and high torque-to-weight ratio; used in vacuum cleaners and power drills.

Pitfall / Pro Tip:

Pure single-phase AC creates a pulsating (standing) magnetic field, not a rotating magnetic field. It requires an auxiliary phase or shading ring to start rotating.

CH07-15 Chapter 7 Level 1 (Core Concept) 📌 AC Machines Slides P.15 & HW7 Problem 3

## Synchronous Motor vs Induction Motor Operation

**QUESTION / EXAM PROBLEM:**

Compare the rotor construction and operating speed of a Synchronous Motor with an Induction Motor. How does a synchronous motor maintain exact synchronous speed?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$N_r = N_s$ (Synchronous) vs $N_r < N_s$ (Induction)

Step-by-Step Solution:

1. Synchronous Motor: - Rotor is excited by an external DC source (or permanent magnets) to establish fixed magnetic poles. - Rotor magnetic poles magnetically lock with the stator's Rotating Magnetic Field (RMF). - Runs at exactly synchronous speed ($N_r = N_s = \frac{120f}{p}$) under all steady-load conditions (zero slip, $s = 0$). 2. Induction Motor: - Rotor currents are induced purely by relative motion between RMF and rotor conductors. - Must always run below synchronous speed ($N_r < N_s$) to produce torque.

Pitfall / Pro Tip:

Synchronous motors can also operate at leading power factor by over-exciting the DC field, functioning as a Synchronous Condenser for power factor correction.

PART 3

# Chapter 8: Transformers

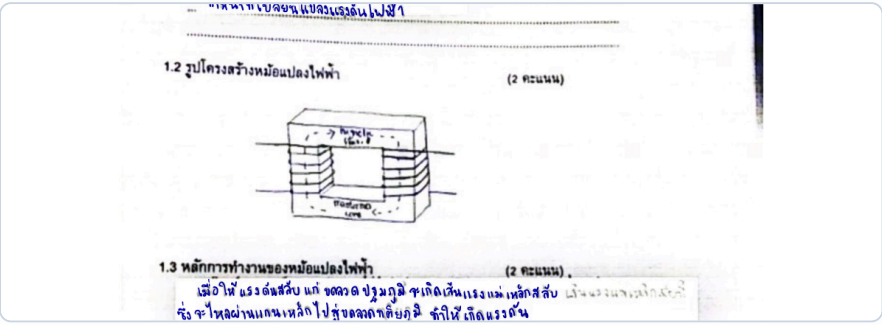
Mutual Induction, Induced EMF Formula, 25 kVA Exam Problem, Audio Matching, Efficiency (11 Flashcards)

**CH08-01** Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.1 & 1.2 (Page 32 - 4 Points)

## Transformer Definition, Purpose & Structure

**QUESTION / EXAM PROBLEM:**

What is an electrical transformer? What is its primary function and structure? Draw/describe its essential construction.




Exam Problem 1.2: Transformer Magnetic Core & Coils

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Static machine transferring AC electrical energy via mutual induction.

Step-by-Step Solution:

1. Definition & Purpose: A static electrical machine that transfers AC electrical power between circuits at the same frequency, stepping voltage up or down to minimize transmission loss or match load requirements. 2. Structure: Consists of: - High-permeability laminated silicon steel core (provides closed magnetic circuit) - Primary winding ($N_1$ turns) connected to input AC source - Secondary winding ($N_2$ turns) connected to output load.

Pitfall / Pro Tip:

A transformer changes voltage and current levels, but NEVER changes power frequency ($f_1 = f_2 = 50$ Hz).

**CH08-02** Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.3 (Page 32 - 4 Points)

## Working Principle & Induced EMF Equation

**QUESTION / EXAM PROBLEM:**

Explain the working principle of a transformer and write Faraday's induced EMF formula including the derivation constant.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$E_{\text{rms}} = 4.44 f N \Phi_m = \sqrt{2}\pi f N \Phi_m$

Step-by-Step Solution:

1. Working Principle: Mutual electromagnetic induction according to Faraday's law. When AC voltage $V_1$ is applied to the primary winding, it drives alternating current that establishes a time-varying magnetic flux $\Phi(t) = \Phi_m \sin(\omega t)$ in the core. This flux links both windings, inducing EMF $e_1 = -N_1 \frac{d\Phi}{dt}$ and $e_2 = -N_2 \frac{d\Phi}{dt}$. 2. RMS EMF Equation: Since $\frac{d}{dt}[\sin(\omega t)] = \omega \cos(\omega t)$, peak value is $E_m = 2\pi f N \Phi_m$. Dividing by $\sqrt{2}$ yields:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N \Phi_m = \sqrt{2}\pi f N \Phi_m = 4.44 f N \Phi_m$$

.

Pitfall / Pro Tip:

Remember the constant 4.44 comes from $\sqrt{2}\pi \approx 1.4142 \times 3.1416 = 4.4428$.

**CH08-03** Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.4 (Page 32 - 2 Points)

## Ideal Transformer 4 Essential Properties

**QUESTION / EXAM PROBLEM:**

List the four fundamental assumptions/properties of an Ideal Transformer.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

1. Zero resistance, 2. Zero leakage, 3. Zero core loss, 4. Infinite permeability

Step-by-Step Solution:

1. Zero winding resistance: $R_1 = 0, R_2 = 0$ (no $I^2R$ copper losses; $V_1 = E_1, V_2 = E_2$). 2. Zero leakage flux: All magnetic flux is completely confined within the core and links 100% of both primary and secondary windings ($L_1 = 0, L_2 = 0$). 3. Zero core loss: No hysteresis loss and no eddy current loss in the core ($P_{\text{core}} = 0$, efficiency $\eta = 100\%$). 4. Infinite core permeability: $\mu_r \to \infty$, meaning zero magnetizing current is required to establish core flux.

Pitfall / Pro Tip:

Direct 2-point exam question! Make sure to list all 4 properties clearly.

**CH08-04** Chapter 8 Level 1 (Core Concept) 📌 Final Exam Problem 1.5 (Page 32 - 4 Points)

## Power Losses in a Practical Transformer

**QUESTION / EXAM PROBLEM:**

What types of power losses occur in a real transformer? Name the 2 primary loss categories and where they occur.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$P_{\text{loss}} = P_{\text{copper}} + P_{\text{core}} = (I_1^2 R_1 + I_2^2 R_2) + (P_h + P_e)$

Step-by-Step Solution:

1. Copper Losses (Winding Losses): Occur in the copper windings due to Joule heating: $P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2$. Variable loss proportional to $(I_{\text{load}})^2$. 2. Core / Iron Losses: Occur in the magnetic core, nearly constant at all load levels: - Hysteresis Loss: Energy lost in continuously reversing magnetic domains ($P_h \propto f B_m^{1.6}$). - Eddy Current Loss: Circulating currents induced in the conductive iron core ($P_e \propto f^2 B_m^2 t^2$). Minimized by using thin laminated sheets insulated by varnish.

Pitfall / Pro Tip:

Core loss depends on voltage and frequency (constant 24/7); copper loss depends on load current squared.

**CH08-05** Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 Final Exam Problem 1.6 (Page 31 - 12 Points)

## Step-Down Transformer Parameter Calculation

**QUESTION / EXAM PROBLEM:**

A single-phase transformer rated at $25\ \text{kVA}, 50\ \text{Hz}$ has primary winding $N_1 = 500\ \text{turns}$ and secondary winding $N_2 = 50\ \text{turns}$. The primary winding is connected to $V_1 = 3,000\ \text{V}$. Calculate: (a) Turns ratio ($a$), (b) Full-load rated primary and secondary currents ( $I_1, I_2$), (c) Induced secondary voltage ($V_2$), (d) Maximum magnetic flux in core ($\Phi_m$).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$a = N_1/N_2\ I_1 = S/V_1, I_2 = S/V_2\ V_2 = V_1/a\ \Phi_m = \frac{V_1}{4.44 f N_1}$

Step-by-Step Solution:

(a) Turns Ratio:

$$a = \frac{N_1}{N_2} = \frac{500}{50} = 10$$

(b) Rated Full-Load Currents:

$$I_1 = \frac{S_{\text{rated}}}{V_1} = \frac{25,000}{3,000} = 8.33\ \text{A}$$
$$I_2 = a \times I_1 = 10 \times 8.33 = 83.33\ \text{A}$$

(c) Secondary Voltage:

$$V_2 = \frac{V_1}{a} = \frac{3,000}{10} = 300\ \text{V}$$

(d) Maximum Magnetic Flux:

$$\Phi_m = \frac{V_1}{4.44 f N_1} = \frac{3,000}{4.44 \times 50 \times 500} = \frac{3,000}{111,000} = 0.0270\ \text{Wb} = 27.0\ \text{mWb}$$

.

Pitfall / Pro Tip:

Exact problem from past exam page 31 (12 points total)! Write all steps and units clearly ($A, V, Wb$).

**CH08-06** Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.8 Problem 2 & Talui Homework Page 4

## 140V/80V Transformer Turns & Currents

**QUESTION / EXAM PROBLEM:**

A single-phase transformer has primary and secondary voltages $V_1 = 140\ \text{V}$ and $V_2 = 80\ \text{V}$. Primary current is $I_1 = 8\ \text{A}$. The secondary winding has $N_2 = 800\ \text{turns}$. Find: (1) Primary winding turns $N_1$, (2) Secondary current $I_2$, (3) Turns ratio $a$.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$a = V_1/V_2 = N_1/N_2\ I_2 = a \times I_1$

Step-by-Step Solution:

1. Turns Ratio:

$$a = \frac{V_1}{V_2} = \frac{140}{80} = 1.75$$

2. Primary Winding Turns:

$$N_1 = a \times N_2 = 1.75 \times 800 = 1,400\ \text{turns}$$

3. Secondary Winding Current:

$$I_2 = a \times I_1 = 1.75 \times 8\ \text{A} = 14.0\ \text{A}$$

.

Pitfall / Pro Tip:

Notice that voltage drops by 1.75 times, so current increases by 1.75 times to conserve total volt-amperes ($S = V_1 I_1 = V_2 I_2 = 1,120\ \text{VA}$ ).

CH08-07 Chapter 8 Level 2 (Past Exam / Calc) HW Ch.8 Problem 3 & Talui Homework Page 4

## 200 kVA, 2200V/220V Transformer Full Calculations

**QUESTION / EXAM PROBLEM:**

A single-phase transformer rated at $200\ \text{kVA}, 2,200\ \text{V}/220\ \text{V}, 50\ \text{Hz}$ has primary winding $N_1 = 200\ \text{turns}$. Find: (1) Rated primary and secondary currents $I_1, I_2$, (2) Secondary winding turns $N_2$, (3) Maximum magnetic flux in core $\Phi_m$.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_1 = S/V_1, I_2 = S/V_2\ a = V_1/V_2 = N_1/N_2\ \Phi_m = \frac{V_1}{4.44fN_1}$

Step-by-Step Solution:

1. Rated Currents:

$$I_1 = \frac{200,000\ \text{VA}}{2,200\ \text{V}} = 90.91\ \text{A}$$

$$I_2 = \frac{200,000\ \text{VA}}{220\ \text{V}} = 909.09\ \text{A}$$

2. Secondary Turns: Turns ratio $a = \frac{2,200}{220} = 10$

$$N_2 = \frac{N_1}{a} = \frac{200}{10} = 20\ \text{turns}$$

3. Maximum Core Flux:

$$\Phi_m = \frac{V_1}{4.44fN_1} = \frac{2,200}{4.44 \times 50 \times 200} = \frac{2,200}{44,400} = 0.0495\ \text{Wb} = 49.5\ \text{mWb}$$

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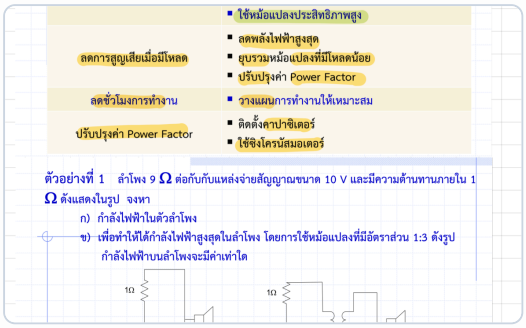
Pitfall / Pro Tip:

Always use rated apparent power in Volt-Amperes ($200\ \text{kVA} = 200,000\ \text{VA}$), not kilo-units.

## Impedance Matching with Transformer for Maximum Power

**QUESTION / EXAM PROBLEM:**

A $10$ V audio source with internal source resistance $R_s = 1\ \Omega$ is connected to a speaker load $R_L = 9\ \Omega$. (1) Find load power without transformer. (2) If an ideal transformer with turns ratio $a = 1/3$ is inserted between source and speaker, find the reflected load resistance and the new load power.




Audio Source Impedance Matching via 1:3 Transformer

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$R'_L = a^2 R_L = \left(\frac{N_1}{N_2}\right)^2 R_L$ $P = I^2 R$

Step-by-Step Solution:

1. Without Transformer:

$$I = \frac{V_s}{R_s + R_L} = \frac{10}{1+9} = 1.0\ \text{A}$$
$$P_{\text{speaker}} = I^2 R_L = (1.0)^2 \times 9 = 9.0\ \text{W}$$

2. With Transformer ($a = 1/3$): Reflected load resistance seen by source:

$$R'_L = a^2 R_L = \left(\frac{1}{3}\right)^2 \times 9 = \frac{1}{9} \times 9 = 1.0\ \Omega$$

Notice $R'_L = R_s = 1.0\ \Omega$ (perfect Maximum Power Transfer match!). New primary current:

$$I_1 = \frac{V_s}{R_s + R'_L} = \frac{10}{1+1} = 5.0\ \text{A}$$

Power delivered to reflected load (and thus speaker, since transformer is ideal):

$$P_{\text{speaker}} = I_1^2 R'_L = (5.0)^2 \times 1.0 = 25.0\ \text{W}$$

Power increased dramatically from $9$ W to $25$ W!

Pitfall / Pro Tip:

Reflected impedance scales as $a^2$: $Z_1 = a^2 Z_2$. In audio systems, transformers match unequal impedances to maximize transferred power.

**CH08-09** Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.8 Problem 4 & Talui Homework Page 5

## DC Voltage Connection to Transformer Primary Hazard

**QUESTION / EXAM PROBLEM:**

What happens if a DC voltage (e.g. 200 V DC) is connected to the primary winding of a transformer? What will the secondary voltage and current be?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$v_2 = -N_2 \frac{d\Phi}{dt} = 0$, $I_1 = \frac{V_{dc}}{R_1}$ (catastrophic overcurrent)

Step-by-Step Solution:

1. Secondary Voltage: $V_2 = 0$ V. Because DC voltage creates static, time-invariant magnetic flux ($\frac{d\Phi}{dt} = 0$), no voltage is induced in the secondary winding. 2. Primary Hazard: On DC, inductive reactance is zero ($X_L = 2\pi f L = 0$ since $f = 0$). There is no counter-EMF to oppose the source. Primary current is limited solely by tiny copper wire resistance $R_1$: $I_1 = V_{\text{dc}}/R_1 \gg I_{\text{rated}}$. The primary winding rapidly overheats and burns out.

Pitfall / Pro Tip:

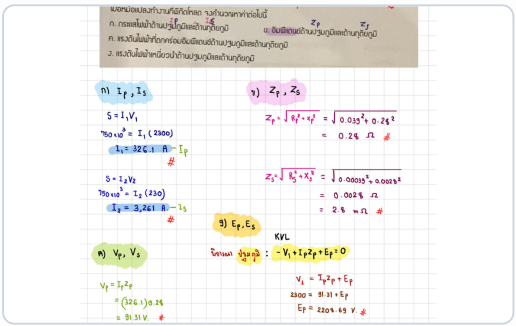
Exam favorite concept! Always state: 1) Secondary output is zero, and 2) Primary winding burns out.

**CH08-10** Chapter 8 🔴 Level 2 (Past Exam / Calc) 📌 FunE-final.pdf Page 10 (Example 5)

## Practical Transformer Equivalent Circuit & Efficiency

**QUESTION / EXAM PROBLEM:**

A $10\ \text{kVA}, 2200/220\ \text{V}, 50\ \text{Hz}$ practical transformer has primary parameters $R_1 = 3.5\ \Omega, X_1 = 6.2\ \Omega$ and secondary parameters $R_2 = 0.035\ \Omega, X_2 = 0.062\ \Omega$. Core loss is $P_{\text{core}} = 150$ W. Find equivalent impedance referred to primary, full-load copper loss, and efficiency at $PF = 1.0$ and $PF = 0.8$ lagging.




Practical Transformer Equivalent Circuit & KVL

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$a = V_1/V_2 = 10$ $R_{eq1} = R_1 + a^2 R_2, X_{eq1} = X_1 + a^2 X_2$ $I_1 = S/V_1$ $P_{\text{cu}} = I_1^2 R_{eq1}$ $\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{cu}} + P_{\text{core}}} \times 100\%$

Step-by-Step Solution:

1. Equivalent Parameters referred to Primary: $a = 2200/220 = 10$ - $R_{eq1} = 3.5 + (10)^2(0.035) = 3.5 + 3.5 = 7.0\ \Omega$ - $X_{eq1} = 6.2 + (10)^2(0.062) = 6.2 + 6.2 = 12.4\ \Omega$ - $Z_{eq1} = 7.0 + j12.4\ \Omega$ 2. Full-Load Primary Current & Copper Loss: $I_1 = 10,000/2,200 = 4.545$ A

$$P_{\text{cu}} = (4.545)^2 \times 7.0 = 144.6\ \text{W}$$

3. Efficiency at Unity PF ($PF = 1.0$): $P_{\text{out}} = 10,000 \times 1.0 = 10,000$ W

$$\eta = \frac{10,000}{10,000 + 144.6 + 150} \times 100\% = \frac{10,000}{10,294.6} \times 100\% = 97.14\%$$

4. Efficiency at $PF = 0.8$ lagging: $P_{\text{out}} = 10,000 \times 0.8 = 8,000$ W

$$\eta = \frac{8,000}{8,000 + 144.6 + 150} \times 100\% = \frac{8,000}{8,294.6} \times 100\% = 96.45\%$$

.

Pitfall / Pro Tip:

Notice how lower power factor degrades transformer efficiency because the real power output drops while $I^2R$ copper loss remains at full rating.

CH08-11 Chapter 8 Level 1 (Core Concept) FunE-final.pdf Page 11 & Lecture Slides

## Energy Conservation in Transformer Operations

**QUESTION / EXAM PROBLEM:**

State three practical methods to save electrical energy in industrial transformer substations.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

1. Reduce no-load loss | 2. Consolidate light loads | 3. Power Factor correction

Step-by-Step Solution:

1. De-energize Unloaded Transformers: Core losses occur 24 hours a day regardless of load. Switching off backup or redundant transformers during weekends/nights eliminates standing core loss. 2. Load Consolidation: Operating two transformers at 20% load is far less efficient than shutting one down and running a single unit at 40-50% load near its peak efficiency point. 3. Power Factor Correction: Installing shunt capacitor banks on the secondary side cancels inductive VARs, reducing secondary line current, which cuts $I^2R$ winding copper losses and lowers transformer operating temperature.

Pitfall / Pro Tip:

Peak efficiency occurs when copper loss equals core loss ($P_{cu} = P_{core}$), which is typically engineered around 50% to 75% rated load.

PART 4

# Chapter 9: Power Transmission Systems & Energy Conservation

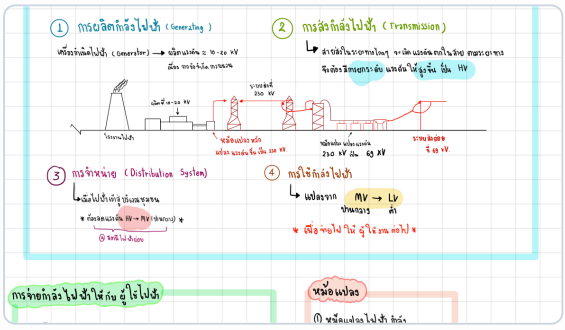
Thai Grid Voltage Stages, EGAT/MEA/PEA, Power Plants, Tower Classifications, Energy Conservation (10 Flashcards)

CH09-01 Chapter 9 Level 1 (Core Concept) Final Exam Problem 5.3 (Page 25 - 4 Points)

## Structure of Electrical Power System & Voltage Levels

**QUESTION / EXAM PROBLEM:**

State the four stages of a power system and classify these standard Thai grid voltages: 500 kV, 22 kV, 380 V, and residential household voltage.




4 Stages of Electric Power Transmission System

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Generation -> Transmission -> Distribution -> Utilization

Step-by-Step Solution:

1. Generation: 10 - 20 kV (alternator terminals) 2. Transmission: 115 kV, 230 kV, 500 kV 3. Distribution: 22 kV, 24 kV, 33 kV 4. Utilization: 380/400 V ($3\phi$) and 220/230 V ($1\phi$) Classifications for Exam Problem 5.3: - 500 kV: Extra High Voltage (EHV) - 22 kV: Medium Voltage (MV) - 380 V: Low / Utilization Voltage - Residential Household Voltage: 220 V (or modern 230 V).

Pitfall / Pro Tip:

Why transmit at Extra High Voltage (500 kV)? Because $P = \sqrt{3}V_L I_L \cos\theta$. Stepping up $V$ reduces line current $I$ proportionally, minimizing $I^2R$ transmission wire loss by $(1/a)^2$.

CH09-02 Chapter 9 Level 1 (Core Concept) 📌 Lecture Slides & Jele Formula Sheet

## Three Electricity Authorities in Thailand & Roles

**QUESTION / EXAM PROBLEM:**

Name the three state electricity authorities in Thailand and state their specific geographic responsibilities.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

EGAT (Generation/Transmission) | MEA (BKK+2) | PEA (74 Provinces)

Step-by-Step Solution:

1. EGAT (Electricity Generating Authority of Thailand / กฟผ.): Responsible for power generation and nationwide high-voltage transmission grid (115-500 kV); sells wholesale power to MEA, PEA, and industrial direct clients. 2. MEA (Metropolitan Electricity Authority / กฟน.): Distributes electricity in 3 central provinces: Bangkok, Nonthaburi, and Samut Prakan. 3. PEA (Provincial Electricity Authority / กฟภ.): Distributes electricity across the remaining 74 provinces in Thailand.

Pitfall / Pro Tip:

Guaranteed exam question! Remember MEA covers strictly Bangkok, Nonthaburi, and Samut Prakan.

CH09-03 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.1 (Page 25 - 2 Points)

## Fuel Mix for Power Generation in Thailand

**QUESTION / EXAM PROBLEM:**

What is the most widely used fuel source for electricity generation in Thailand, and what fuel source is used least?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Most: Natural Gas (~55-60%) | Least: Fuel Oil / Heavy Fuel Oil

Step-by-Step Solution:

- Most used: Natural Gas (ก๊าซธรรมชาติ) accounts for over 55-60% of total electrical energy in Thailand (Gulf of Thailand, Myanmar, and imported LNG). - Followed by: Coal / Lignite (Mae Moh power plant) and Renewable/Hydro imports from Lao PDR. - Least used: Heavy fuel oil (น้ำมันเตา) and diesel, due to extremely high operating costs and environmental impact.

Pitfall / Pro Tip:

Do not answer solar or oil as the main source; it is overwhelmingly Natural Gas.

CH09-04 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.2 (Page 25 - 4 Points)

## Base Load vs Peak Load Power Plants & Startup Times

**QUESTION / EXAM PROBLEM:**

Compare Base Load and Peak Load power plants. Which plant type has the shortest startup time, and which has the longest?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Base Load: Steam/Coal (6-8 hrs) | Peak Load: Hydro (<5 mins)

Step-by-Step Solution:

1. Base Load Plants: Run continuously 24/7 to supply minimum baseline grid demand. High capital cost, low running cost. - Longest Startup Time: Steam thermal power plants (e.g. Mae Moh coal/lignite) take 6 - 8 hours to bring boilers up to operating pressure from cold start. 2. Peak Load Plants: Dispatched quickly to handle rapid spikes in daily demand. - Shortest Startup Time: Hydroelectric power plants (โรงไฟฟ้าพลังน้ำ) can synchronize and deliver full load in less than 5 minutes. Gas turbines take 10 - 15 minutes.

Pitfall / Pro Tip:

Direct 4-part exam question: Shortest = Hydro; Longest = Steam; Base = Steam/Coal; Peak = Hydro/Gas turbine.

**CH09-05** Chapter 9 Level 1 (Core Concept) FunE-final.pdf Page 14 & Lecture Slides

## Transmission Tower Classifications & Structures

**QUESTION / EXAM PROBLEM:**

Describe the structural features and conductor configurations used on Thai grid transmission towers at 500 kV, 230 kV, and 115 kV.



Transmission Tower Classifications (500 kV, 230 kV, 115 kV, 69 kV)

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

500 kV (Quadruple bundle) | 230 kV (Twin bundle) | 115 kV (Single conductor)

Step-by-Step Solution:

1. 500 kV Towers: Massive steel lattice towers. Use Quadruple Bundle Conductors (4 sub-conductors spaced per phase) to expand effective conductor diameter, suppressing Corona discharge and Audible noise, with ~28-35 insulator discs per string. 2. 230 kV Towers: Double-circuit lattice towers using Twin Bundle Conductors (2 sub-conductors per phase) with ~14-17 insulator discs. 3. 115 kV Towers: Single-conductor lattice or spun concrete poles with ~7-9 insulator discs, linking sub-transmission substations to city borders.

Pitfall / Pro Tip:

Bundle conductors are used at 500 kV and 230 kV primarily to reduce high-voltage Corona discharge loss and radio interference.

**CH09-06** Chapter 9 Level 1 (Core Concept) FunE-final.pdf Page 13 & PEA/MEA Regulations

## Customer Categories & Supply Voltage Thresholds

**QUESTION / EXAM PROBLEM:**

How are electricity consumers categorized by supply voltage in Thailand? State the voltage levels supplied to: (1) Residential households, (2) Medium business (30-999 kVA), (3) Large industrial (>10,000 kVA).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

<30 kVA: LV 230/400 V | 30-999 kVA: MV 22/33 kV | >10 MVA: HV 115 kV

Step-by-Step Solution:

1. Low Voltage Customers (< 30 kVA): - Residential households and small shops. - Supplied at $230\ \mathrm{V}$ ($1\phi$, 2-wire) or $400/230\ \mathrm{V}$ ($3\phi$, 4-wire). 2. Medium Voltage Customers (30 - 999 kVA): - Commercial buildings, hotels, small factories. - Supplied at $22\ \mathrm{kV}$ or $33\ \mathrm{kV}$ (PEA) / $12\ \mathrm{kV}$ or $24\ \mathrm{kV}$ (MEA) with customer-owned step-down transformer. 3. High Voltage Customers (> 10,000 kVA / 10 MVA): - Heavy chemical plants, steel mills, large industrial estates. - Supplied directly from $115\ \mathrm{kV}$ or $230\ \mathrm{kV}$ grid via private dedicated substation.

Pitfall / Pro Tip:

Large customers buy power at high voltage because tariff rates per unit (kWh) are cheaper, and transmission loss over long industrial site feeders is minimized.

**CH09-07** Chapter 9 Level 1 (Core Concept) 📌 FunEE Final Note P.5 & Lecture Slides

## Power Plant Specifics in Thailand (Bhumibol, Ubolratana, Mae Moh, Nam Pong)

**QUESTION / EXAM PROBLEM:**

Match the following major Thai power plants to their generation type, location, and key features: (1) Bhumibol Dam, (2) Ubolratana Dam, (3) Mae Moh, (4) Nam Pong.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Bhumibol: Ping River / Tak | Ubolratana: Khon Kaen | Mae Moh: Lignite / Lampang | Nam Pong: Combined Cycle / Khon Kaen

Step-by-Step Solution:

1. Bhumibol Dam (เขื่อนภูมิพล): - Location: Tak Province, Ping River. - Thailand's first multipurpose concrete arch dam; major hydro peak-load plant. 2. Ubolratana Dam (เขื่อนอุบลรัตน์): - Location: Khon Kaen Province, Pong River. - First hydroelectric dam in Northeastern Thailand (Isan). 3. Mae Moh Power Plant (โรงไฟฟ้าแม่เมาะ): - Location: Lampang Province. - Thailand's largest coal/lignite-fired steam thermal base-load power plant ($2,625\ \mathrm{MW}$). 4. Nam Pong Power Plant (โรงไฟฟ้าน้ำพอง): - Location: Khon Kaen Province ($32\ \mathrm{km}$ from downtown KKU). - Combined-cycle power plant fueled by domestic natural gas from the Sinphuhorm/Nam Phong field.

Pitfall / Pro Tip:

Very relevant for KKU engineering students! Ubolratana Dam and Nam Pong power plant are located directly in Khon Kaen.

**CH09-08** Chapter 9 Level 1 (Core Concept) 📌 FunEE Final Note P.6-7 & Lecture Slides

## Residential Solar PV Rooftop System Components

**QUESTION / EXAM PROBLEM:**

Draw the flow path and name the 5 essential components of a grid-connected residential Solar PV system.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

PV Modules -> DC/DC Charge Controller -> Battery -> DC/AC Inverter -> Fusebox / Bi-directional Meter

Step-by-Step Solution:

1. PV Solar Modules (Panels): Convert solar irradiance into DC electrical power. 2. DC/DC Converter / Charge Controller: Regulates charging voltage and extracts maximum power (MPPT). 3. Battery Storage: Stores electrical energy in DC form for night/emergency backup. 4. DC/AC Power Inverter: Converts direct current (DC) into synchronized $230\ \mathrm{V}, 50\ \mathrm{Hz}$ alternating current (AC). 5. Consumer Unit (Fusebox) & Bi-directional Net Meter: Distributes power to household loads and exports excess power to the MEA/PEA grid.

Pitfall / Pro Tip:

Solar cells produce pure DC; household appliances run on AC. The Inverter is the crucial bridging component.

**CH09-09** Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.5 (Page 24 - 2 Points)

## Energy Conservation in Household Washing Machines

**QUESTION / EXAM PROBLEM:**

State two practical, effective methods to conserve electrical energy when operating a household washing machine.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

1. Wash full rated loads | 2. Cold water wash | 3. Air dry clothes

Step-by-Step Solution:

1. Wash with Full Rated Load: Consolidate dirty laundry to run at maximum rated drum capacity rather than running multiple partial loads (each cycle consumes roughly the same mechanical and pumping energy). 2. Use Cold Water Wash: Avoid electrical water heating cycles. In washing machines with built-in heaters, heating water accounts for 80% to 90% of total electrical energy per wash cycle. 3. Spin and Air Dry: Utilize high-speed mechanical spin extraction and sun/air drying instead of electric heated tumble dryers.

Pitfall / Pro Tip:

Direct 2-point exam question from page 24!

CH09-10 Chapter 9 Level 1 (Core Concept) 📌 Final Exam Problem 5.6 (Page 24 - 3 Points)

### Energy Conservation in Industrial Manufacturing Plants

**QUESTION / EXAM PROBLEM:**

State three practical engineering methods to reduce electrical energy consumption and costs in an industrial manufacturing plant.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

1. Load scheduling (TOU) | 2. Capacitor banks (PF) | 3. Transformer management

Step-by-Step Solution:

1. Peak Load Shifting (TOU Tariff Optimization): Schedule energy-intensive machinery (electric arc furnaces, heavy crushers, pumping) during Off-Peak tariff hours (22:00 - 09:00 and weekends), dramatically lowering demand charges. 2. Power Factor Correction with Capacitor Banks: Maintain overall plant power factor above $0.95$ lagging to eliminate utility reactive power penalty fees and reduce line current $I^2R$ heat dissipation in plant distribution feeders. 3. Transformer Management & High-Efficiency Motors: Consolidate lightly loaded transformers during slow shifts to eliminate no-load core losses, and upgrade old motors to Premium Efficiency (IE3/IE4) with Variable Speed Drives (VFDs).

Pitfall / Pro Tip:

Direct 3-point exam question from page 24!

PART 5

## Chapter 10: Measuring Instruments & Sensors

PMMC Principles, DC Ammeters & Voltmeters with CDRX, Ayrton Shunts, Ohmmeters, Wattmeters (14 Flashcards)

CH10-01 Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.1 (Page 30 - 4 Points)

### PMMC / Galvanometer Working Principle & Deflection Torque

**QUESTION / EXAM PROBLEM:**

Explain the working principle of a PMMC (Permanent Magnet Moving Coil) instrument. State the deflecting torque formula and deflection angle relation.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$T_d = BANI = BILdN$ $T_c = K\theta \implies \theta = \frac{BAN}{K}I$

Step-by-Step Solution:

1. Principle: Motor principle (Lorentz force). When current $I$ flows through the moving coil situated in the permanent magnetic field, a deflecting torque $T_d$ is exerted on the coil sides. 2. Deflecting Torque: $T_d = BANI$, where $B$ = magnetic flux density (T), $A$ = coil area ( $m^2$), $N$ = turns, $I$ = current (A). 3. Controlling Torque: Opposed by hair springs: $T_c = K\theta$. 4. Steady-State Deflection: At balance ($T_d = T_c$ ):

$$\theta = \left(\frac{BAN}{K}\right) I$$

Deflection $\theta$ is linearly proportional to current $I$, giving PMMC a perfectly UNIFORM (linear) scale.

Pitfall / Pro Tip:

PMMC responds strictly to DC average current. If pure sinusoidal AC is applied, average torque is zero and pointer merely vibrates around zero.

**CH10-02** Chapter 10 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.10 Problem 1 & Talui Homework Page 6

## PMMC Torque & Deflection Calculation

**QUESTION / EXAM PROBLEM:**

A moving-coil instrument has $N = 150$ turns, coil area $A = 100\ \text{mm}^2$, air gap flux density $B = 0.15\ \text{T}$, and current $I = 5\ \text{mA}$. Spring constant $K = 2 \times 10^{-6}\ \text{N} \cdot \text{m/deg}$. Calculate: (1) Deflecting torque ($T_d$), (2) Deflection angle ($\theta$ in degrees and radians).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$T_d = NBAI\ \theta = T_d/K$

Step-by-Step Solution:

1. Deflecting Torque: $A = 100\ \text{mm}^2 = 100 \times 10^{-6}\ \text{m}^2 = 10^{-4}\ \text{m}^2$

$$T_d = 150 \times 0.15 \times (100 \times 10^{-6}) \times (5 \times 10^{-3}) = 1.125 \times 10^{-5}\ \text{N} \cdot \text{m}$$

2. Deflection Angle:

$$\theta = \frac{T_d}{K} = \frac{1.125 \times 10^{-5}}{2 \times 10^{-6}} = 5.625^\circ$$

In radians:

$$\theta = 5.625 \times \frac{\pi}{180} = 0.0982\ \text{rad}$$

.

Pitfall / Pro Tip:

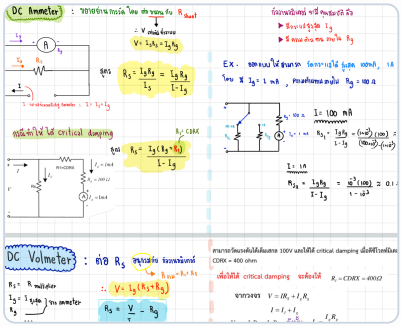
Always convert coil area from $\text{mm}^2$ to $\text{m}^2$ by multiplying by $10^{-6}$! Never forget this conversion.

**CH10-03** Chapter 10 🔴 Level 2 (Past Exam / Calc) 📌 HW Ch.10 Problem 2 & Talui Homework Page 6

## DC Ammeter Design with and without CDRX

**QUESTION / EXAM PROBLEM:**

Design a DC ammeter with full scale $I = 100\ \text{mA}$ using a galvanometer with $I_g = 10\ \text{mA}$ and $R_g = 50\ \Omega$. Calculate shunt resistor $R_s$ : (a) Without damping resistor, (b) With critical damping resistor CDRX = $400\ \Omega$ connected in series with coil.




DC Ammeter with Shunt and CDRX Wiring

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

(a) $R_s = \frac{I_g R_g}{I - I_g}$ (b) $R_s = \frac{I_g(R_g + \text{CDRX})}{I - I_g}$

Step-by-Step Solution:

(a) Without CDRX:

$$R_s = \frac{10\ \text{mA} \times 50\ \Omega}{100\ \text{mA} - 10\ \text{mA}} = \frac{0.5\ \text{V}}{0.09\ \text{A}} = 5.56\ \Omega$$

(b) With CDRX = $400\ \Omega$ (connected in series with $R_g$): Total meter branch resistance $R_{\text{branch}} = R_g + \text{CDRX} = 50 + 400 = 450\ \Omega$

$$R_s = \frac{I_g(R_g + \text{CDRX})}{I - I_g} = \frac{10\ \text{mA} \times 450\ \Omega}{90\ \text{mA}} = \frac{4.5\ \text{V}}{0.09\ \text{A}} = 50.0\ \Omega$$

.

Pitfall / Pro Tip:

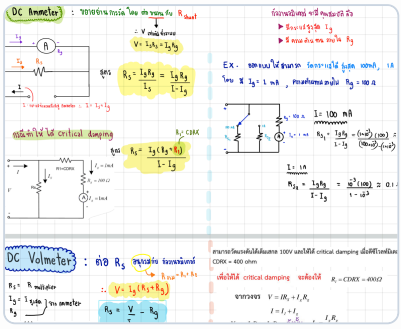
In a DC ammeter, CDRX is connected in SERIES with the moving coil inside the meter branch; the shunt $R_s$ is in parallel with that entire branch.

CH10-04 Chapter 10 Level 2 (Past Exam / Calc) HW Ch.10 Problem 3 & Talui Homework Page 7

## DC Voltmeter Design with and without CDRX

**QUESTION / EXAM PROBLEM:**

Design a DC voltmeter with full scale $V = 50$ V using a galvanometer with $I_g = 10$ mA and $R_g = 50\ \Omega$. Calculate series multiplier $R_s$: (a) Without damping resistor, (b) With critical damping resistor CDRX = $400\ \Omega$ connected in parallel across the coil.




DC Voltmeter with Series Multiplier and Parallel CDRX

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

(a) $R_s = \frac{V}{I_g} - R_g$ (b) $R_s = \frac{V - I_g R_g}{I_g + I_{\text{cdrx}}}$

Step-by-Step Solution:

(a) Without CDRX:

$$R_s = \frac{V}{I_g} - R_g = \frac{50}{0.010} - 50 = 5,000 - 50 = 4,950\ \Omega$$

(b) With CDRX = $400\ \Omega$ (connected in PARALLEL across the coil): - Voltage across coil at FSD: $V_g = I_g R_g = 10\ \text{mA} \times 50\ \Omega = 0.5$ V - Current through CDRX: $I_{\text{cdrx}} = V_g / \text{CDRX} = 0.5/400 = 1.25$ mA - Total circuit current: $I_{\text{tot}} = I_g + I_{\text{cdrx}} = 10\ \text{mA} + 1.25\ \text{mA} = 11.25$ mA - Required series multiplier:

$$R_s = \frac{V - V_g}{I_{\text{tot}}} = \frac{50 - 0.5}{0.01125} = \frac{49.5}{0.01125} = 4,400\ \Omega$$

.

Pitfall / Pro Tip:

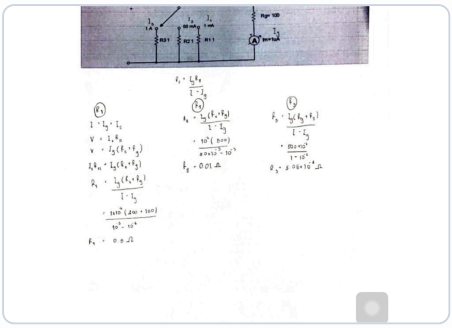
In a voltmeter, CDRX is placed in PARALLEL with the coil to damp oscillation, so the series resistor must supply both coil current and CDRX current!

**CH10-05** Chapter 10 🔴 Level 2 (Past Exam / Calc) 📌 Final Exam Problem 3.5 (Page 29 - 12 Points)

## Multi-Range DC Ammeter Design with CDRX

**QUESTION / EXAM PROBLEM:**

Design a multi-range DC ammeter with ranges $I_1 = 1\ \text{mA}$, $I_2 = 50\ \text{mA}$, and $I_3 = 1\ \text{A}$ using a galvanometer with internal resistance $R_g = 100\ \Omega$, full-scale deflection current $I_g = 1\ \mu\text{A}$, and damping resistor $\text{CDRX} = 400\ \Omega$ in series with coil. Find shunt resistors $R_1, R_2, R_3$.



Exam Problem 3.5: Multi-Range DC Ammeter with CDRX Circuit

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$R_{\text{branch}} = R_g + \text{CDRX} = 500\ \Omega\ V_{\text{branch}} = I_g \times R_{\text{branch}} = 1\ \mu\text{A} \times 500\ \Omega = 500\ \mu\text{V} = 5 \times 10^{-4}\ \text{V}\ R_s = \frac{V_{\text{branch}}}{I - I_g}$

Step-by-Step Solution:

Total meter branch resistance: $R_{\text{branch}} = 100 + 400 = 500\ \Omega$. Branch voltage at FSD: $V_m = 1 \times 10^{-6}\ \text{A} \times 500\ \Omega = 5 \times 10^{-4}\ \text{V}\ (0.5\ \text{mV})$. 1. Range 1 ($I = 1\ \text{mA} = 10^{-3}\ \text{A}$):

$$R_1 = \frac{5 \times 10^{-4}\ \text{V}}{10^{-3} - 10^{-6}} \approx \frac{5 \times 10^{-4}}{10^{-3}} = 0.5\ \Omega$$

2. Range 2 ($I = 50\ \text{mA} = 50 \times 10^{-3}\ \text{A}$): R_2 = \frac{5 \times 10^{-4}\text{ V}}{50 \times 10^{-3}} = 0.01\ \Omega$ 3. Range 3 ($I = 1\text{ A}$): R_3 = \frac{5 \times 10^{-4}\text{ V}}{1\text{ A}} = 5 \times 10^{-4}\ \Omega = 0.0005\ \Omega = 0.5\text{ m}\Omega$$.

Pitfall / Pro Tip:

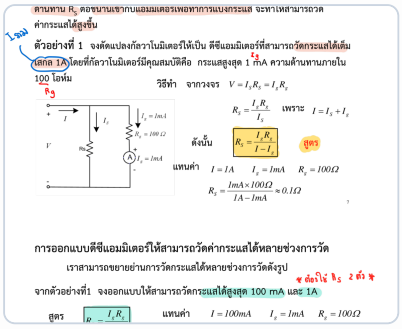
Exact 12-point exam problem! As measurement range increases, the required shunt resistance decreases proportionally.

CH10-06 Chapter 10 Level 2 (Past Exam / Calc) FunE-final.pdf Page 18 & Slide P.12

## Ayrton Shunt (Universal Shunt) for Multi-Range Ammeter

**QUESTION / EXAM PROBLEM:**

What is an Ayrton (Universal) Shunt in a multi-range DC ammeter? Why is it superior to individual switch shunts?




Individual Shunts vs Ayrton Universal Shunt Circuit

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Series chain of resistors: $R_1 + R_2 + R_3$

Step-by-Step Solution:

1. Structure: Employs a continuous series string of tapped resistors ($R_1, R_2, R_3$) permanently connected across the meter movement. 2. Decisive Advantage over Individual Shunts: When switching between ranges, rotary switch contacts momentarily break. With individual shunts, the meter coil is momentarily left unshunted and exposed to the full line current, burning out the delicate hair spring. With an Ayrton shunt, the meter is NEVER left open or unshunted during range switching.

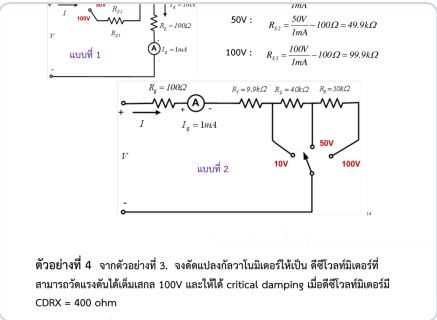
Pitfall / Pro Tip:

Always mention the safety feature: the meter coil is never open-circuited while switching current ranges.

## Multi-Range Voltmeter (Individual vs Tapped Series)

**QUESTION / EXAM PROBLEM:**

Compare Individual Multipliers versus Series Tapped Multipliers for designing a multi-range DC voltmeter. State the resistance formulas for ranges $V_1, V_2, V_3$.




ตัวอย่างที่ 4 จากตัวอย่างที่ 3. จงดัดแปลงกัลวาโนมิเตอร์ให้เป็น ดีซีโวลท์มิเตอร์ที่สามารถวัดแรงดันได้เต็มเสกล 100V และให้ได้ critical damping เมื่อดีซีโวลท์มิเตอร์มี CDRX = 400 ohm

Multi-Range Voltmeter: Individual vs Series Tapped Multipliers

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Tapped: $R_1 = \frac{V_1}{I_g} - R_g, R_2 = \frac{V_2 - V_1}{I_g}, R_3 = \frac{V_3 - V_2}{I_g}$

Step-by-Step Solution:

1. Method 1: Individual Multipliers (Each range uses a separate dedicated resistor):

$$R_{s1} = \frac{V_1}{I_g} - R_g, \quad R_{s2} = \frac{V_2}{I_g} - R_g, \quad R_{s3} = \frac{V_3}{I_g} - R_g$$

2. Method 2: Series Tapped Multipliers (Resistors connected in cumulative series chain): - Range 1 ($V_1$): $R_1 = \frac{V_1}{I_g} - R_g$ - Range 2 ($V_2$): $R_2 = \frac{V_2 - V_1}{I_g}$ - Range 3 ($V_3$): $R_3 = \frac{V_3 - V_2}{I_g}$ Total series resistance for range 3 is $R_1 + R_2 + R_3$. Benefit of Tapped Series: Saves board space and requires standard lower resistor values.

Pitfall / Pro Tip:

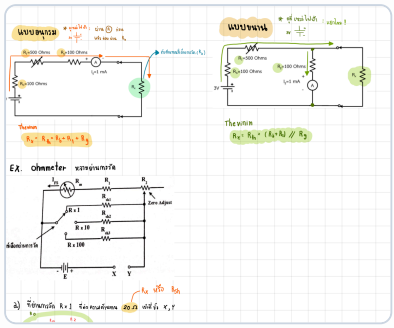
For tapped series, remember each resistor handles only the INCREMENTAL voltage $\Delta V = V_k - V_{k-1}$, not the total voltage from zero.

**CH10-08** Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

## Series-Type Ohmmeter Working Principle & Scale

**QUESTION / EXAM PROBLEM:**

Explain the operating principle of a Series-Type Ohmmeter. Describe its zero-calibration, open-circuit state, and why its scale is reversed and non-linear.



Series-Type vs Shunt-Type Ohmmeter Circuits

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_m = \frac{E}{R_1+R_2+R_g+R_x}$ $R_h = R_{\text{in}} = R_1 + R_2 + R_g$

Step-by-Step Solution:

1. Schematic: Battery $E$, current limiting resistor $R_1$, zero-adjust rheostat $R_2$, galvanometer $R_g$, and unknown resistor $R_x$ in series. 2. Zero Calibration ($R_x = 0$, leads shorted): Full-scale deflection current flows: $I_{\text{fsd}} = \frac{E}{R_{\text{in}}}$. Pointer deflects to FAR RIGHT ($0\ \Omega$). 3. Open Circuit ($R_x = \infty$, leads open): Zero current flows: $I = 0$. Pointer stays at FAR LEFT ($\infty\ \Omega$). 4. Half-Scale Deflection ($I_m = 0.5I_{\text{fsd}}$): Occurs exactly when $R_x = R_{\text{in}} = R_h$. Half-scale resistance equals the meter's internal circuit resistance. 5. Non-linear Scale: Current is inversely proportional to $R_x$, causing severe crowding at the left (high resistance) end.

Pitfall / Pro Tip:

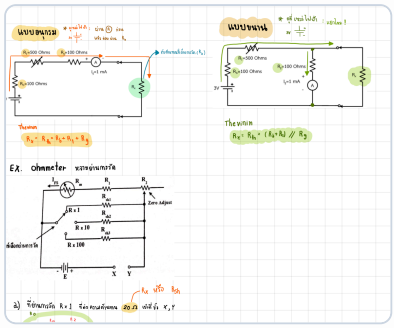
Series ohmmeter scale runs BACKWARDS: $0\ \Omega$ is at the right, and $\infty\ \Omega$ is at the left!

**CH10-09** Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

## Shunt-Type Ohmmeter for Low Resistance Measurement

**QUESTION / EXAM PROBLEM:**

Explain the operating principle of a Shunt-Type Ohmmeter. Why is its scale direction opposite to a series ohmmeter, and what resistance range is it suited for?



Shunt Ohmmeter Schematic

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Scale: $0\ \Omega$ at LEFT, $\infty\ \Omega$ at RIGHT

Step-by-Step Solution:

1. Structure: The unknown resistor $R_x$ is connected in PARALLEL (shunt) across the galvanometer movement, with a switch $S$ to disconnect the internal battery when not measuring. 2. Zero State ($R_x = 0$): $R_x$ completely shorts out the galvanometer coil. Meter current is zero ($I_m = 0$). Pointer stays at FAR LEFT ($0\ \Omega$). 3. Open Circuit ($R_x = \infty$): No current is shunted away; full current flows through the meter coil ($I_m = I_{\text{fsd}}$). Pointer deflects to FAR RIGHT ($\infty\ \Omega$). 4. Application: Specifically suited for measuring very LOW resistances ( $0.5\ \Omega - 100\ \Omega$), such as motor windings and transformer coils.

Pitfall / Pro Tip:

Remember: Series ohmmeter has $0\ \Omega$ on RIGHT; Shunt ohmmeter has $0\ \Omega$ on LEFT (same direction as voltmeter/ammeter).

**CH10-10** Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 17 & Lecture Slides

## Multi-Range Ohmmeter & Zero-Adjust Rheostat Function

**QUESTION / EXAM PROBLEM:**

In a multi-range analog multimeter (ohmmeter), why is a Zero-Adjust Rheostat ($R_{\text{adj}}$) necessary, and why must the user re-zero when switching ranges ($R \times 1, R \times 10, R \times 1\text{k}$)?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Compensates for battery aging: $E \downarrow$ and $r_{\text{bat}} \uparrow$

Step-by-Step Solution:

1. Battery Aging Compensation: As internal chemical batteries are used and age, their open-circuit terminal voltage gradually decreases ($E \downarrow$) and their internal resistance increases ($r_b \uparrow$). Without adjustment, full-scale deflection cannot be reached when shorting leads. 2. Zero-Adjust Rheostat Function: By turning the thumbwheel rheostat, the user varies internal resistance so that $I_{\text{fsd}} = E/R_{\text{in}}$ is precisely restored at $R_x = 0$. 3. Range Switching: Different ranges ($R \times 1, R \times 10, R \times 100$) switch in different shunt and series resistors, altering total internal circuit resistance $R_{\text{in}}$. Therefore, the meter MUST be re-zeroed every time a new range is selected.

Pitfall / Pro Tip:

Always short the test leads and adjust to $0\ \Omega$ before making any resistance measurement!

**CH10-11** Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.2 (Page 30 - 2 Points)

## Rectifier AC Voltmeter & Form Factor

**QUESTION / EXAM PROBLEM:**

What is the operating principle of a Rectifier AC Voltmeter, and how does it relate average rectified DC to the indicated RMS value?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_{\text{rms}} = 1.11 \times V_{\text{avg}}$ (for pure sine waves)

Step-by-Step Solution:

1. Principle: Uses semiconductor diodes (half-wave or full-wave bridge rectifier) to convert an alternating voltage into unidirectional pulsating DC, which is then measured by a standard PMMC movement. 2. Scale Calibration: PMMC inherently responds to average value ($V_{\text{avg}} = \frac{2}{\pi}V_m = 0.636V_m$). The meter scale is multiplied by the Sine Wave Form Factor:

$$\text{Form Factor} = \frac{V_{\text{rms}}}{V_{\text{avg}}} = \frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$$

3. Reading Distortion: If the measured waveform is not a pure sine wave (e.g. square, triangle, or inverter PWM harmonics), the reading will have significant waveform error.

Pitfall / Pro Tip:

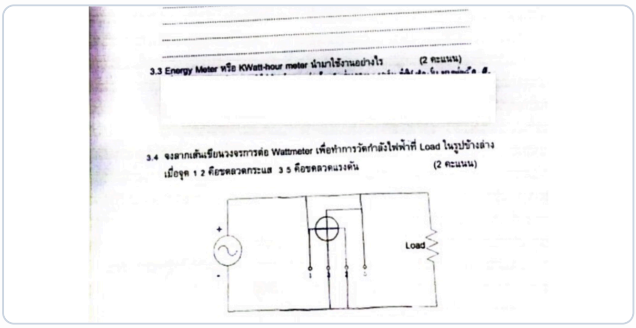
A rectifier meter reads accurately ONLY for pure sine waves. For distorted waveforms, a True-RMS meter is required.

**CH10-12** Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.4 (Page 30 - 4 Points)

## Wattmeter Connection to Load (Current vs Potential Coils)

**QUESTION / EXAM PROBLEM:**

In an electrodynamometer Wattmeter, how must the Current Coil and Potential (Voltage) Coil be connected into the circuit to measure load power?




Exam Problem 3.4: Wattmeter Terminal Connections (1-2 and 3-5)

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Current Coil (1-2) in SERIES | Potential Coil (3-5) in PARALLEL

Step-by-Step Solution:

1. Current Coil (Terminals 1 - 2): - Made of thick copper wire with very low resistance. - Must be connected in SERIES with the load to carry full load current ($I_{\text{load}}$). 2. Potential / Voltage Coil (Terminals 3 - 5): - Made of many turns of fine wire in series with a high non-inductive resistor. - Must be connected in PARALLEL across the load to sense load voltage ($V_{\text{load}}$). 3. Deflection Torque:

$$T_d \propto V_{\text{load}} I_{\text{load}} \cos\theta = P \quad [\text{W}]$$

.

Pitfall / Pro Tip:

Never connect the current coil in parallel across the supply—it has virtually zero resistance and will cause an explosive short circuit!

**CH10-13** Chapter 10 Level 1 (Core Concept) 📌 Final Exam Problem 3.3 (Page 30 - 2 Points)

## Energy Meter (Kilowatt-Hour Meter) Function

**QUESTION / EXAM PROBLEM:**

What is an induction-type Energy Meter (Watt-hour meter) and what physical quantity and unit does it measure?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$\text{Energy} = \int P(t)dt \quad [\text{kW} \cdot \text{h}]$

Step-by-Step Solution:

1. Purpose: An integrating instrument that measures total electrical energy consumed over a period of time, commonly used by MEA/PEA for customer billing. 2. Quantity & Unit: Measures active electrical energy in units of Kilowatt-hours ($\text{kW} \cdot \text{h}$ or 'units'). 3. Principle: Eddy current induction motor principle driving a rotating aluminum disk geared to a register counter.

Pitfall / Pro Tip:

A wattmeter measures instantaneous Power ($W$), while an energy meter measures cumulative Energy ($W \cdot h$ or $kW \cdot h$). Do not confuse the two!

**CH10-14** Chapter 10 Level 1 (Core Concept) 📌 FunE-final.pdf Page 15 & Slide P.3-5

## Instrument Accuracy, Precision, and Error Formulas

**QUESTION / EXAM PROBLEM:**

Define and write the mathematical formulas for: (1) Absolute Error, (2) Relative Error (%), (3) Relative Accuracy, (4) Precision in electrical measurements.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$\epsilon = |Y_n - X_n| \; \%\text{Error} = \frac{|Y_n - X_n|}{Y_n} \times 100\% \; \%\text{Accuracy} = 100\% - \%\text{Error} \; \text{Precision} = 1 - \left|\frac{X_n - \bar{X}_n}{\bar{X}_n}\right|$

Step-by-Step Solution:

1. Absolute Error ($\epsilon$): Difference between true/expected value ($Y_n$) and measured value ($X_n$):

$$\epsilon = |Y_n - X_n|$$

2. Percent Relative Error ($\epsilon_r$):

$$\%\text{Error} = \frac{|Y_n - X_n|}{Y_n} \times 100\%$$

3. Percent Accuracy ($A$):

$$\%\text{Accuracy} = \left(1 - \frac{|Y_n - X_n|}{Y_n}\right) \times 100\% = 100\% - \%\text{Error}$$

4. Precision: Measure of repeatability/consistency across multiple readings:

$$\text{Precision} = 1 - \left|\frac{X_n - \bar{X}_n}{\bar{X}_n}\right|$$

where $\bar{X}_n$ is the arithmetic average of all repeated measurements.

Pitfall / Pro Tip:

High precision does NOT guarantee high accuracy! A miscalibrated instrument can give highly repeatable (precise) readings that are completely wrong (inaccurate).

PART 6

# Chapter 11 / Supplement: Semiconductor & Electronic Devices

Diodes, Knee Voltages, LEDs, Solar PV Cells, BJT Terminal Currents, MOSFETs, Op-Amps (12 Flashcards)

**SEMI-01** Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.a (Page 24 - 2 Points)

## Intrinsic vs Extrinsic, N-Type vs P-Type Semiconductors

**QUESTION / EXAM PROBLEM:**

Explain the difference between Intrinsic and Extrinsic semiconductors. Compare N-type and P-type semiconductors in terms of dopants and majority carriers.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

N-type: Group V dopant (electrons) | P-type: Group III dopant (holes)

Step-by-Step Solution:

1. Intrinsic Semiconductor: Chemically pure semiconductor (e.g. pure Silicon Si, Germanium Ge) with equal concentrations of free electrons and holes ($n = p = n_i$). Very low electrical conductivity at room temperature. 2. Extrinsic Semiconductor: Doped with impurity atoms to dramatically enhance conductivity: - N-type: Pure Si doped with Pentavalent atoms (Group V: Phosphorus P, Arsenic As, Antimony Sb with 5 valence electrons). Majority carriers = Electrons; Minority carriers = Holes. - P-type: Pure Si doped with Trivalent atoms (Group III: Boron B, Gallium Ga, Indium In with 3 valence electrons). Majority carriers = Holes; Minority carriers = Electrons.

Pitfall / Pro Tip:

Exact 2-point exam question from page 24! Memory aid: P in Pentavalent gives Negative electrons; T in Trivalent gives Positive holes.

**SEMI-02** Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.b (Page 24 - 2 Points)

## MOSFET Definition, Properties & Uses

**QUESTION / EXAM PROBLEM:**

What is a MOSFET? State its essential electrical characteristics and its primary application/advantage in electronic circuits.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Metal-Oxide-Semiconductor Field-Effect Transistor (Voltage-controlled)

Step-by-Step Solution:

1. Definition: Metal-Oxide-Semiconductor Field-Effect Transistor, a unipolar transistor where current transport involves only majority carriers. 2. Characteristics: Voltage-controlled device with extremely high input impedance ($10^{10} - 10^{15}\ \Omega$) due to the insulating $SiO_2$ gate oxide layer. Gate draws virtually zero DC current ($I_G \approx 0$). 3. Application / Benefit: Uses gate electric field to modulate and control current flow between Drain and Source. Widely used for high-speed switching, microprocessors (CMOS), and power converters.

Pitfall / Pro Tip:

Exact question from page 24: State that it is a unipolar transistor whose output current is controlled by an electric field / voltage.

**SEMI-03** Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4.c (Page 24 - 1 Point)

## Integrated Circuit (IC) Definition & Significance

**QUESTION / EXAM PROBLEM:**

What is an Integrated Circuit (IC)? What is its main function and significance in modern electronics?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Microchip integrating transistors, diodes, resistors on single silicon die

Step-by-Step Solution:

1. Definition: A miniaturized electronic circuit consisting of thousands to billions of interconnected semiconductor devices (transistors, diodes, resistors, capacitors) fabricated on a single tiny substrate of semiconductor material (silicon wafer). 2. Significance: Enables massive reduction in physical size, weight, and power consumption, while offering dramatic improvements in reliability, operating speed, and mass-production cost.

Pitfall / Pro Tip:

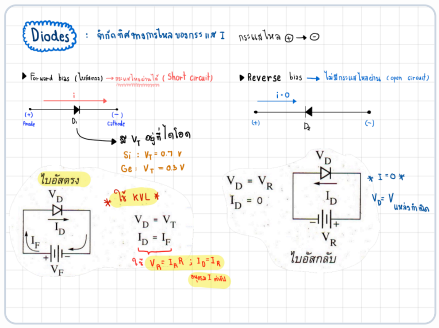
Exact 1-point question from past exam page 24.

**SEMI-04** Semiconductors Level 1 (Core Concept) 📌 Lecture Slides Ch.11 & Jele Formula Sheet

## PN Junction Diode Biasing & Knee Voltages

**QUESTION / EXAM PROBLEM:**

Compare Forward Bias and Reverse Bias in a PN junction diode. State the barrier potential (knee voltage) values for Silicon (Si) and Germanium (Ge).




PN Diode Biasing, Forward Conduction and LED Circuit

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Forward: P to +, N to - (conduction) | Reverse: P to -, N to + (blocking) $V_k(\text{Si}) = 0.7\ \text{V}, V_k(\text{Ge}) = 0.3\ \text{V}$

Step-by-Step Solution:

1. Forward Bias: External positive connected to P-side (anode), negative to N-side (cathode). Overcomes internal barrier, depletion width shrinks, and heavy forward current flows once applied voltage exceeds knee voltage. 2. Reverse Bias: Positive to N-side, negative to P-side. Depletion layer widens, blocking current except for tiny temperature-dependent reverse saturation leakage current ($I_0$). 3. Knee Voltages: - Silicon (Si): $V_{\text{knee}} = 0.7\ \text{V}$ - Germanium (Ge): $V_{\text{knee}} = 0.3\ \text{V}$.

Pitfall / Pro Tip:

In almost all circuit analysis, Silicon diode forward voltage drop is taken as 0.7 V.

**SEMI-05** Semiconductors 🔴 Level 2 (Past Exam / Calc) 📌 FunE-final.pdf Page 22 & Lecture Slides

## Diode Shockley Equation, Dynamic Resistance & LED Circuit

**QUESTION / EXAM PROBLEM:**

1. Write the Shockley diode equation and define thermal voltage $V_T$ at room temperature (25°C). 2. For an LED circuit with $V_{CC} = 5\ \text{V}$, LED knee voltage $V_{AK} = 2.0\ \text{V}$, find the required current limiting resistor $R$ to maintain $I_D = 15\ \text{mA}$.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_D = I_S(e^{V_D/nV_T} - 1)$ $V_T = 25\ \text{mV}$ $R = \frac{V_{CC}-V_{AK}}{I_D}$

Step-by-Step Solution:

1. Shockley Diode Equation:

$$I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$$

where $I_S$ is reverse saturation current, $n$ is ideality factor ($1 \le n \le 2$), and thermal voltage $V_T = \frac{kT}{q} \approx 25\ \text{mV}$ (or $26\ \text{mV}$) at room temperature ($298\ \text{K}$). 2. Dynamic / Small-Signal Resistance:

$$r_d = \frac{nV_T}{I_D} = \frac{25\ \text{mV}}{I_D}$$

3. LED Limiting Resistor Calculation:

$$R = \frac{V_{CC} - V_{AK}}{I_D} = \frac{5.0 - 2.0\ \text{V}}{15 \times 10^{-3}\ \text{A}} = \frac{3.0}{0.015} = 200\ \Omega$$

.

Pitfall / Pro Tip:

Never connect an LED directly to a voltage supply without a series resistor; without current limiting, it will burn out instantaneously.

**SEMI-06** Semiconductors 🔴 Level 2 (Past Exam / Calc) 📌 Lecture Slides Ch.11 & Jele Formula Sheet

## Zener Diode Voltage Regulator Operation

**QUESTION / EXAM PROBLEM:**

Explain how a Zener Diode regulates voltage. A $10\ \text{V}$ Zener diode is connected across a variable input voltage $V_{\text{in}} = 15\ \text{V}$ with a series resistor $R_s = 200\ \Omega$ and load resistance $R_L = 1\ \text{k}\Omega$. Calculate Zener current $I_Z$.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_s = \frac{V_{\text{in}}-V_Z}{R_s}$ $I_L = V_Z/R_L$ $I_Z = I_s - I_L$

Step-by-Step Solution:

1. Principle: Operates in the REVERSE BREAKDOWN region where voltage remains virtually constant ($V_Z$) across wide variations in reverse current. 2. Calculation: - Regulated load voltage: $V_L = V_Z = 10\ \text{V}$ - Source branch current: $I_s = \frac{V_{\text{in}}-V_Z}{R_s} = \frac{15-10}{200} = \frac{5}{200} = 25\ \text{mA}$ - Load branch current: $I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{1{,}000\ \Omega} = 10\ \text{mA}$ - Zener diode current: $I_Z = I_s - I_L = 25\ \text{mA} - 10\ \text{mA} = 15\ \text{mA}$.

Pitfall / Pro Tip:

For regulation to hold, $I_Z$ must remain greater than knee current $I_{Z(min)}$ and less than maximum power rating current $I_{Z(max)} = P_Z/V_Z$.

**SEMI-07** Semiconductors Level 1 (Core Concept) 📌 FunE-final.pdf Page 22 & Lecture Slides

## Solar Photovoltaic Cell Equivalent Circuit & Parameters

**QUESTION / EXAM PROBLEM:**

Draw the equivalent circuit of a photovoltaic solar cell and define: Open-circuit voltage ($V_{oc}$), Short-circuit current ($I_{sc}$), Maximum power point ($P_{mp}$), and Fill Factor (FF).

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$\mathrm{FF} = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}} = \frac{P_{\max}}{V_{oc} I_{sc}}$

Step-by-Step Solution:

1. Equivalent Circuit: Light-generated constant current source $I_L$ in parallel with a forward-biased PN junction diode $D$, parallel shunt leakage resistance $R_{sh}$, and series contact resistance $R_s$. 2. Key Parameters: - $V_{oc}$: Open-circuit terminal voltage under illumination with zero load current (typically $0.6 - 0.7$ V per silicon cell). - $I_{sc}$: Short-circuit current when terminals are connected together ($V = 0$). - $P_{\max} = V_{mp} \times I_{mp}$: Point on I-V curve where power output is maximized. - Fill Factor (FF): Ratio measuring the squareness of the I-V curve: $\mathrm{FF} = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}$, typically $0.75 - 0.85$ for high-quality silicon solar cells.

Pitfall / Pro Tip:

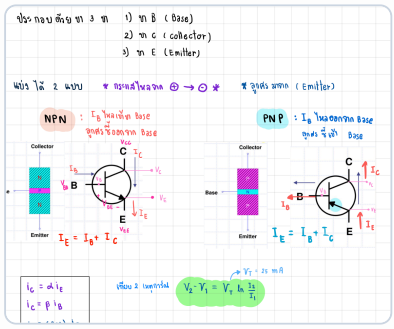
A single silicon solar cell produces only ~0.6 V, so 60 to 72 cells are connected in series inside a panel to generate usable voltages (30-40 V).

**SEMI-08** Semiconductors Level 1 (Core Concept) 📌 Final Exam Problem 4 & Lecture Slides

## BJT Transistor Terminal Currents & Gain Relations

**QUESTION / EXAM PROBLEM:**

State the fundamental KCL current relationship between Emitter ($I_e$), Collector ($I_c$), and Base ($I_b$) currents in a BJT, and the relationships with current gains $\beta$ and $\alpha$.




BJT NPN/PNP Symbols, Terminal Currents and MOSFET Structure

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$I_e = I_c + I_b \; I_c = \beta I_b \; \alpha = \frac{\beta}{\beta+1} = \frac{I_c}{I_e}$

Step-by-Step Solution:

1. Terminal Current Law (KCL): Emitter current is the sum of collector and base currents:

$$I_e = I_c + I_b$$

2. Common-Emitter DC Current Gain ($\beta$ or $h_{FE}$):

$$I_c = \beta I_b$$

Typically $\beta$ ranges from $50$ to $300$. 3. Common-Base DC Current Gain ($\alpha$):

$$\alpha = \frac{I_c}{I_e} = \frac{\beta}{\beta + 1}$$

Typically $\alpha \approx 0.98 - 0.998$ (slightly less than 1.0).

Pitfall / Pro Tip:

Because base current $I_b$ is tiny (typically $\mu$A), $I_e \approx I_c$ in many rough approximations, but NEVER forget $I_e = I_c + I_b$ in exact KCL equations.

**SEMI-09** Semiconductors Level 1 (Core Concept) 📌 FunE-final.pdf Page 23 & Lecture Slides

## BJT Transistor as an Electronic Switch

**QUESTION / EXAM PROBLEM:**

Explain how a BJT operates as an electronic switch. What conditions place the transistor into Cutoff (Switch OFF) versus Saturation (Switch ON)?

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Cutoff: $I_B = 0, V_{CE} = V_{CC}$ | Saturation: $I_B \geq I_{C(\text{sat})}/\beta, V_{CE} \approx 0.2\ \text{V}$

Step-by-Step Solution:

1. Cutoff Region (Switch OPEN / OFF): - Condition: Base-Emitter voltage $V_{BE} < 0.7\ \text{V}$ (zero base drive). - Currents: $I_B = 0, I_C = 0$. - Output Voltage: $V_{CE} = V_{CC}$ (full supply voltage across open collector-emitter terminals). 2. Saturation Region (Switch CLOSED / ON): - Condition: Base is heavily overdriven with $I_B \geq \frac{I_{C(\text{sat})}}{\beta}$, where $I_{C(\text{sat})} \approx \frac{V_{CC}}{R_C}$. - Currents: Both junctions are forward-biased, collector current reaches maximum limited by load. - Output Voltage: $V_{CE} = V_{CE(\text{sat})} \approx 0.2\ \text{V}$ (acts like a closed switch with minimal drop).

Pitfall / Pro Tip:

When using a transistor as a switch, always ensure $I_B > I_{C(sat)}/\beta$ (overdrive factor of 2 to 3) to guarantee deep saturation.

**SEMI-10** Semiconductors Level 1 (Core Concept) 📌 Ch10IntroSemiDevices.pdf Slide P.21

## BJT vs FET / MOSFET Detailed Comparison

**QUESTION / EXAM PROBLEM:**

Provide a comprehensive comparison between BJT (Bipolar Junction Transistor) and FET / MOSFET across 6 electrical parameters.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

BJT (Current-controlled, Bipolar) vs FET (Voltage-controlled, Unipolar)

Step-by-Step Solution:

Direct exam comparison table from Slide 21: 1. Control Mechanism: - BJT: Current-controlled device ($I_c = \beta I_b$). - FET/MOSFET: Voltage-controlled device ($I_D = g_m V_{GS}$). 2. Input Impedance: - BJT: Low ($1\ \text{k}\Omega - 10\ \text{k}\Omega$). - FET/MOSFET: Extremely high ($10^{10} - 10^{15}\ \Omega$, draws zero gate current). 3. Charge Carriers: - BJT: Bipolar (conduction uses both electrons and holes). - FET: Unipolar (conduction uses only majority carriers in the channel). 4. Noise: - BJT: More noisy. - FET: Less noisy. 5. Thermal Stability: - BJT: Less temperature stable (prone to thermal runaway). - FET: More temperature stable. 6. Physical Size on Chip: - BJT: Larger size. - MOSFET: Much smaller size (ideal for billion-transistor IC microchips).

Pitfall / Pro Tip:

Memorize this exact comparison table from Slide 21! It appears regularly in theory short-answer and multiple-choice questions.

**SEMI-11** Semiconductors Level 2 (Past Exam / Calc) X-Class Review Page 7-9 & Lecture Slides

## Half-Wave vs Full-Wave Rectifiers & PIV Rating

**QUESTION / EXAM PROBLEM:**

Compare Half-Wave Rectifier, Full-Wave Center-Tapped Rectifier, and Full-Wave Bridge Rectifier in terms of: (1) Number of diodes, (2) DC average output voltage ($V_{\text{dc}}$), (3) Peak Inverse Voltage (PIV) rating across each diode.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

Half-wave: $V_{\text{dc}} = \frac{V_m}{\pi} = 0.318V_m$, $\text{PIV} = V_m$ Full-wave: $V_{\text{dc}} = \frac{2V_m}{\pi} = 0.636V_m$ Center-Tapped PIV = $2V_m$ | Bridge PIV = $V_m$

Step-by-Step Solution:

1. Half-Wave Rectifier: - 1 diode - $V_{\text{dc}} = V_m/\pi = 0.318V_m$ - Ripple frequency = $f_{\text{in}}$ ($50\ \text{Hz}$) - $\text{PIV} = V_m$. 2. Full-Wave Center-Tapped Rectifier: - 2 diodes - $V_{\text{dc}} = 2V_m/\pi = 0.636V_m$ - Ripple frequency = $2f_{\text{in}}$ ($100\ \text{Hz}$) - $\text{PIV} = 2V_m$ (Diodes must withstand twice peak voltage!). 3. Full-Wave Bridge Rectifier: - 4 diodes - $V_{\text{dc}} = 2V_m/\pi = 0.636V_m$ - Ripple frequency = $2f_{\text{in}}$ ($100\ \text{Hz}$) - $\text{PIV} = V_m$ (Does NOT require expensive center-tapped transformer).

Pitfall / Pro Tip:

Exam multiple choice favorite: Full-wave center-tapped diode requires $\text{PIV} = 2V_m$, whereas Bridge diode requires only $\text{PIV} = V_m$.

**SEMI-12** Semiconductors Level 1 (Core Concept) Ch10IntroSemiDevices.pdf Slide P.23-24

## Operational Amplifier (Op-Amp 741) Characteristics

**QUESTION / EXAM PROBLEM:**

What is an Operational Amplifier (Op-Amp)? Identify its 5 primary terminal pins and state the 3 ideal Op-Amp properties.

**FORMULA, SOLUTION & PRO TIPS:**

Formula / Principle:

$V_{\text{out}} = A_{OL}(V_+ - V_-)$

Step-by-Step Solution:

1. Definition: A high-gain DC-coupled differential voltage amplifier integrated circuit (e.g. standard $\mu\text{A}741$). 2. Terminal Pins: - Pin 2: Inverting input ($V_-$) - Pin 3: Non-inverting input ($V_+$) - Pin 6: Output ($V_{\text{out}}$) - Pin 7: Positive power supply ($V_S^+$) - Pin 4: Negative power supply ($V_S^-$). 3. Ideal Op-Amp Properties: - Infinite open-loop voltage gain: $A_{OL} = \infty$ - Infinite input impedance: $R_{\text{in}} = \infty$ (no input current, $I_+ = I_- = 0$) - Zero output impedance: $R_{\text{out}} = 0$ - Infinite bandwidth and zero input offset voltage.

Pitfall / Pro Tip:

Virtual ground concept: When negative feedback is present, the Op-Amp drives the differential input to zero ($V_+ = V_-$).